NAAC GRADE A+
ACCREDITED UNIVERSITY

REVA UNIVERSITY
Bengaluru, India

School of Computing and Information Technology

B.Tech – Computer Science and Information Technology)

First Year Handbook 2025-26

RUKMINI EDUCATIONAL
Charitable Trust

www.reva.edu.in

CONTENTS

Sl. No	Particulars	Page No.
1	Chancellor's Message	3
2	Pro-Chancellor's Message	5
3	Vice- Chancellor's Message	7
4	Rukmini Educational Charitable Trust	8
5	About REVA University	9
5.1	Vision, Mission, Objectives of REVA University	10
6	Director's Message	11
7	About the School of COMPUTING AND INFORMATION TECHNOLOGY	12
7.1	Program Overview	13
7.2	Program Educational Objectives (PEOs)	15
7.3	Program Outcomes (POs)	15
7.4	Program Specific Outcomes (PSOs)	16
8	Academic Regulations 2025-26	17-36
9	Curriculum Structure for B. Tech Computer Science and Information Technology	37-38
10	Detailed Syllabus	39-107

Chancellor’s Message

“Education is the most powerful weapon which you can use to change the world.”

- Nelson Mandela.

At REVA University, we are firm believers in this truth.
The power of education lies not only in the acquisition of knowledge but in its ability to transform lives, communities, and nations. As the world rapidly evolves, driven by technological advancements and global interconnectedness, education remains the catalyst for progress, innovation, and positive change.

Today, we live in an era where knowledge is no longer confined to books or classrooms. Information is accessible at the touch of a button, and opportunities for learning abound in every corner of the world. Yet, the true essence of education lies beyond the mere accumulation of information. It is about cultivating critical thinking, fostering creativity, and empowering individuals to contribute meaningfully to society. At REVA University, we are committed to nurturing seekers of knowledge who aspire to make a difference in the world.

Guided by our founding philosophy of "Knowledge is Power," we strive to create an environment where intellectual curiosity is encouraged, and dreams are transformed into reality. India is a land of immense talent and potential, and it is our duty as educators to provide the spark that ignites this potential. Through the transformative power of education, we aim to shape future leaders who possess not only technical proficiency but also strong ethical values and a commitment to social responsibility.

A university is more than just a place of learning; it is a place of growth, exploration, and transformation. Our faculty, with their expertise and dedication, are at the heart of this transformation. They are more than teachers; they are mentors who guide students on their journey of self-discovery and academic excellence. Our student- centric, transformational approach ensures that every learner is given the opportunity to explore their full potential and exceed their own expectations.

At REVA University, we take great pride in our state-of-the-art infrastructure and facilities, designed to provide an inspiring and conducive environment for both academic and extracurricular pursuits. Our campus is a vibrant space where students are encouraged to challenge their minds, develop their skills, and grow as individuals.

As we move forward, I am reminded of the words of Benjamin Disraeli: "A university should be a place of light, of liberty, and of learning." This vision continues to inspire us at REVA University, where we work as a team to create a brighter future for our students and our society.

I invite you to join us on this journey of enlightenment, growth, and transformation. Together, let us lay the foundation for a future built on values, wisdom, and knowledge.

Dr. P. Shyama Raju
Chancellor
REVA University

Pro Chancellor's Message

REVA University has emerged as a premier destination for higher education across diverse fields such as engineering, science, commerce, management, architecture, law, arts, and humanities. Our commitment to excellence in education is reinforced by the adoption of cutting-edge technologies and innovative teaching methods that ensure our students are equipped for the future.

The integration of modern tools and ICT-based technologies is at the core of our academic philosophy. We focus on digital learning, project-based learning, and personalized learning experiences that cater to individual student needs. By harnessing the power of advanced technologies such as AI-powered learning platforms, data analytics, and virtual/augmented reality, we are able to offer dynamic, interactive educational experiences that transcend traditional classroom boundaries. This technological transformation enables us to deliver STEM education more effectively while providing our faculty with ongoing professional development to stay at the forefront of teaching innovations.

Our programs are meticulously designed after a thorough analysis of current industry needs and trends, with a focus on knowledge assimilation, practical application, and global employability. We recognize the importance of preparing students not just for today's job market but for a rapidly changing future where automation, artificial intelligence, and data-driven decision-making will play pivotal roles. To meet these demands, we emphasize hands-on learning, skill development, and innovation, ensuring that our students are well-prepared to thrive in their respective fields.

At REVA University, we have implemented the Choice Based Credit System and Continuous Assessment Grading Pattern (CBCS – CAGP) in all our programs. This system provides students with the flexibility to choose subjects that align with their interests while developing essential skills. CBCS courses integrate knowledge on local, regional, national, and global issues, fostering a comprehensive understanding of the world and enabling students to become entrepreneurial and employable in a competitive global marketplace. Furthermore, students are offered a variety of value-added courses to enhance their skillsets, ensuring they are equipped to navigate the evolving demands of their chosen careers.

The future of the engineering profession, and indeed many other fields, will be shaped by dramatic technological and societal changes. The rise of automation, sustainability concerns, and increased globalization will create both opportunities and challenges for the next generation of professionals. REVA University is fully prepared to meet these challenges head-on. Through state-of-the-art laboratories, research centers, and partnerships with premier industries and academic institutions, we are committed to creating talented professionals and leaders who can navigate these future transformations.

Our growth and success have been built on a foundation of excellence in curriculum design, student-centric teaching methods, and hands-on learning practices. I extend my gratitude to our students, parents, faculty, staff, and well-wishers for their contributions in helping REVA University become a next-generation, globally recognized education hub.

Mr. Umesh S. Raju
Pro Chancellor
REVA University

Vice Chancellor's Message

The last two decades have seen remarkable growth in higher education in India and across the globe. The move towards interdisciplinary studies and interactive learning has opened up several options as well as created multiple challenges.

A strong believer and practitioner of the dictum "Knowledge is Power", REVA University has been on the path of delivering quality education by developing young human resources on the foundation of ethical and moral values while boosting their leadership qualities, research culture and innovative skills.

Built over 50 acres of green campus, this 'temple of learning' has excellent and state-of-the-art infrastructure facilities conducive to a higher teaching-learning environment and research. The main objective of the University is to provide higher education of global standards and hence, all the programmes are designed to meet international standards. Highly experienced and qualified faculty members, continuously engaged in the maintenance and enhancement of a student-centric learning environment through innovative pedagogy, form the backbone of the University. All the programmes offered by REVA University follow the Choice Based Credit System (CBCS) with Outcome Based Approach.

The flexibility in the curriculum has been designed with industry-specific goals in mind and the educator enjoys complete freedom to appropriate the syllabus by incorporating the latest knowledge and stimulating the creative minds of the students. Benchmarked with the course of studies of various institutions of repute, our curriculum is extremely contemporary and is a culmination of efforts of great think tanks - a large number of faculty members, experts from industries and research-level organizations. The evaluation mechanism employs continuous assessment with grade point averages. We believe sincerely that it will meet the aspirations of all stakeholders- students, parents and the employers of the graduates and postgraduates of REVA University.

At REVA University, research, consultancy and innovation are regarded as our pillars of success. Most of the faculty members of the University are involved in research by attracting funded projects from various research-level organizations like STI HUB, ISRO, DST, VGST, DBT, DRDO, AICTE and industries. The outcome of the research is passed on to students through live projects from industries. The entrepreneurial zeal of the students is encouraged and nurtured through EDPs and EACs. With firm faith in the saying, "Intelligence plus character –that is the goal of education" (Martin Luther King, Jr.), I strongly believe REVA University is marching ahead in the right direction, providing a holistic education to the future generation and playing a positive role in nation building. We reiterate our commitment to providing premium quality education accessible to all and an environment for the growth of overall personality development leading to generating "GLOBAL PROFESSIONALS". Welcome to the portals of REVA University.

Dr. Sanjay R. Chitnis

Vice Chancellor, REVA University

Rukmini Educational Charitable Trust

It was the dream of late Smt. Rukmini Shyama Raju to impart education to millions of underprivileged children as she knew the importance of education in the contemporary society. The dream of Smt. Rukmini Shyama Raju came true with the establishment of Rukmini Educational Charitable Trust (RECT), in the year 2002. **Rukmini Educational Charitable Trust** (RECT) is a Public Charitable Trust, set up in 2002 with the objective of promoting, establishing and conducting academic activities in the fields of Arts, Architecture, Commerce, Education, Engineering, Environmental Science, Legal Studies, Management and Science & Technology, among others. In furtherance of these objectives, the Trust has set up the REVA Group of Educational Institutions comprising of REVA Institute of Technology & Management (RITM), REVA Institute of Science and Management (RISM), REVA Institute of Management Studies (RIMS), REVA Institute of Education (RIE), REVA First Grade College (RFGC), REVA Independent PU College at Kattigenahalli, Ganganagar and Sanjaynagar and now REVA University. Through these institutions, the Trust seeks to fulfill its vision of providing world class education and create abundant opportunities for the youth of this nation to excel in the areas of Arts, Architecture, Commerce, Education, Engineering, Environmental Science, Legal Studies, Management and Science & Technology.

Every great human enterprise is powered by the vision of one or more extraordinary individuals and is sustained by the people who derive their motivation from the founders. The Chairman of the Trust is Dr. P. Shyama Raju, a developer and builder of repute, a captain of the industry in his own right and the Chairman and Managing Director of the DivyaSree Group of companies. The idea of creating these top notched educational institutions was born of the philanthropic instincts of Dr. P. Shyama Raju to do public good, quite in keeping with his support to other socially relevant charities such as maintaining the Richmond road park, building and donating a police station, gifting assets to organizations providing accident and trauma care, to name a few.

The Rukmini Educational Charitable Trust drives with the main aim to help students who are in pursuit of quality education for life. REVA is today a family of ten institutions providing education from PU to Post Graduation and Research leading to PhD degrees. REVA has well qualified experienced teaching faculty of whom majority are doctorates. The faculty is supported by committed administrative and technical staff. Over 13,000 students study various courses across REVA’s three campuses equipped with exemplary state-of-the-art infrastructure and conducive environment for the knowledge driven community.

About REVA University

REVA University is a State Private University established in Karnataka State under the Government of Karnataka Act No. 13 in the year 2012 in Bengaluru, the IT capital of India. REVA University, recognised by the University Grants Commission (UGC) and approved by the All India Council for Technical Education (AICTE), has an A+ grade from NAAC. The University has a sprawling green campus spread over 43 acres of land.

The University has a DIAMOND Band ranking from QS I Gauge. As per QS Asian University Rankings, it is ranked 47th among all the private Universities of India and 6th among all private universities of Karnataka. In less than a decade, REVA University, Bengaluru, has established itself as a Global University in education by earning recognition as a forward-thinking institution across all disciplines.

The University currently offers 38 full-time undergraduate Programmes, 33 full-time postgraduate programmes, 20 Ph.D. programmes, and certification and diploma programmes. The University offers programmes under the faculty of Engineering, Architecture, Science and Technology, Commerce, Management Studies, Law, Arts & Humanities, and Performing Arts & Indic Studies. REVA offers some of the trending fields of study as undergraduate courses in Sports Science (B. Sc.), Agricultural Engineering (B. Tech), and Aerospace Engineering (B. Tech) which are full-time application-based programmes with a unique blend of theory and practical components.

With state-of-the-art infrastructure, the University has created a vibrant academic environment conducive to higher learning and research. This includes 200 smart classrooms that support blended learning, real-industry-like labs that foster on-the-job learning in students, a tech-enabled library with over 1 lakh collections of books, and most importantly, modern pedagogy. REVA currently has numerous students on campus from around the country. The campus has exclusive Halls of Residence which provides comfortable accommodation for boys and girls, apart from catering to the needs of all cuisine and ensuring adequate amenities are provided to make their stay an extended home.

In its mission to become a social impact university, REVA University has initiated several Corporate Social Responsibility initiatives. Jagruti, Abhivriddhi, Vanamahotsava, Education on Wheels, and Pragna are a few of the several projects in REVA. REVA has now moved on to become a Social Impact University and has aligned with the United Nations Sustainable Development Goals. Through these initiatives, REVA aspires to become an innovative University by developing a social connection with leadership qualities, ethical and moral values, research culture, and innovative skills through higher education of global standards.

Vision of REVA University

To become a technologically advanced, sustainable global university dedicated to the wellbeing of all.

Mission of REVA University

- Provide learner-centric education leveraged with cutting edge technologies.
- Foster stewardship by nurturing talent, leadership qualities, and entrepreneurial thinking in a safe and secure environment.
- Promote liberal studies and foster the pursuit of performing arts, literature, sports, and other creative and intellectual disciplines.
- Promote a culture of collaboration and cooperation.
- Serve humanity and promote sustainability through higher education based on universal values

Objectives of REVA University

- Developing a sense of ethics in the University and community, making it conscious of its obligations to society and the nation.
- Performing all the functions of interest to its major constituents like faculty, staff, students, and the society to reach a leadership position.
- Smooth transition from teacher-centric focus to learner-centric processes and activities.
- To offer high-quality education in a competitive manner.
- Creation, preservation and dissemination of knowledge and attainment of excellence in different disciplines.

Director Message

I congratulate and welcome all the students to the esteemed school of Computing and Information Technology (C & IT)). You are in the right campus to become a computer technocrat. The rising needs of automation in Industry 4.0 and improvising living standards have enabled rapid development of computer software and hardware technologies. Thus providing scope and opportunity to generate more human resources in the areas of computers and IT. The B.Tech, M.Tech and Ph.D. programs offered in the school are designed to cater the requirements of industry and society. The curriculum is designed meticulously in association with persons from industries (TCS, CISCO, AMD, MPHASIS, etc.), academia and research organizations (IISc, IIIT, Florida University, Missouri S & T University, etc.). The Curriculum caters to local, national, regional and global developmental needs. Maximum number of courses are integrated with cross cutting issues relevant to professional ethics, global needs, human values, environment and sustainability. The courses in the curriculum focus on skill development, innovation and entrepreneurship.

This handbook presents the B.Tech program offered by CSIT . The program is of 4 years duration and split into 8 semesters. The courses are classified into foundation core, hard core, and soft core courses. Hard core courses represent fundamentals study requirements of B.Tech CSIT program. Soft courses provide flexibility to students to choose the options among several courses as per the specialization, such as, Artificial Intelligence, Fuzzy Logic and Systems, Cognitive science, and predictive analytics etc. Theoretical foundations of engineering, science, and Information Science are taught in first two and half years. Later, advanced courses and recent technologies are introduced in subsequent semesters for pursuing specialization.

The important features of the B.Tech programs offered by the school are as follows:

1) Choice based course selection and teacher selection, 2) Studies in emerging areas like Machine Learning, Artificial Intelligence, Data Analytics, Cloud Computing, Python/R Programming, NLP, IoT and Cloud security, 3) Short and long duration Internships 4) Opportunity to pursue MOOC course as per the interest in foundation and soft core courses, 5) Attain global and skill certification as per the area of specialization, 6) Self-learning components, 7) Experiential, practice, practical, hackathons, and project based learning, 8) Mini projects and major projects with research orientation and publication, 9) Soft skills training and 10) Platform for exhibiting skills in cultural, sports and technical activities through clubs and societies.

The school has well qualified faculty members in the various areas of computing and IT including cloud computing, security, IOT, AI, ML and DL, software engineering, computer networks, information technology, cognitive computing, block chain technology etc. State of art laboratories are available for the purpose of academics and research.

Director, School of Computing and Information Technology

About School of Computing and Information Technology

The school has a rich blend of experienced and committed faculty who are well-qualified in various aspects of computing and information technology apart from the numerous state-of-the-art digital classrooms and laboratories having modern computing equipment. The School offers 3 full-time undergraduate programs, B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning), B.Tech in Computer Science and Information Technology, B.Tech in Information Science and Engineering and the following two postgraduate programs: M.Tech in Artificial Intelligence and M.Tech in Cybersecurity. In addition, the school has a research centre in which students can conduct cutting edge research leading to a PhD degree. Curriculum of both undergraduate and postgraduate programs have been designed through a collaboration of academic and industry experts to bridge the growing gap between industry and academia. This makes the program highly practical-oriented, and thus industry-resilient. The B.Tech programs aims to create quality human resources to play leading roles in the contemporary, competitive industrial and corporate world. The masters' degrees focus on quality research and design in the core and application areas of Artificial Intelligence and Information Technology to foster a sustainable world and to enhance the global quality of life by adopting enhanced design techniques and applications. This thought is reflected in the various courses offered in the masters' programs.

School Vision

To produce excellent quality technologists and researchers of global standards in computing and Information technology who have potential to contribute to the development of the nation and the society with their expertise, skills, innovative problem-solving abilities, strong moral and ethical values.

School Mission

- To create state of the art computing labs infrastructure and research facilities in information technology.
- To provide student-centric learning environment in Computing and Information technology through innovative pedagogy and education reforms.
- To encourage research, innovation and entrepreneurship in computing and information technology through industry/academia collaborations and extension activities
- Organize programs through club activities for knowledge enhancement in thrust areas of information technology.
- To enhance leadership qualities among the youth and enrich personality traits, promote patriotism.

B. Tech Computer Science and Information Technology

Program Overview

Computer Science and Information Technology (CS & IT) encompasses a variety of topics that relates to computation and applications of computing like, development of algorithms, analysis of algorithms, programming languages, software design, computer hardware, e-commerce, business information technology, Data Analytics, Machine Learning, Block Chain Technology, Augmented Virtual Reality, Mobile Application Development, IoT, Wireless Sensor network, Web Technology. Computer Science and Information Technology (CS & IT) has roots in Electrical Engineering, Mathematics, and Linguistics. In the past Computer Science and information science were taught as part of mathematics or engineering departments and in the last 3 decades they are emerged as separate engineering fields. In the present information era

(Knowledge era), the Computer Science and information technology program will see an exponential growth as the future machines work on artificial intelligence.

The oldest known complex computing device, called the Antikythera mechanism, dates back to 87 B.C., to calculate astronomical positions and help Greeks navigate through the seas. Computing took another leap in 1843, when English mathematician Ada Lovelace wrote the first computer algorithm, in collaboration with Charles Babbage, who devised a theory of the first programmable computer. But the modern computing-machine era began with Alan Turing's conception of the Turing Machine and three Bell Labs scientists invention of the transistor, which made modern-style computing possible, and landed them the 1956 Nobel Prize in Physics. For decades, Computing Technology was exclusive to the government and the military; later, academic institutions came online, and Steve Wozniak built the circuit board for Apple-1, making home computing practicable. On the connectivity side, Tim Berners-Lee created the World Wide Web, and Marc Andreessen built a browser, and that's how we came to live in a world where our glasses can tell us what we're looking at. With wearable computers, embeddable chips, smart appliances, and other advances in progress and on the horizon, the journey towards building smarter, faster and more capable computers is clearly just beginning. Computers have become ubiquitous part of modern life, and new applications are introduced every day. The use of computer technologies is also commonplace in all types of organizations, in academia, research, industry, government, private and business organizations. As computers become even more pervasive, the potential for computer-related careers will continue to grow and the career paths in computer-related fields will become more diverse. Since 2001, global information and communication technologies (ICTs) have become more powerful, more accessible, and more widespread. They are now pivotal in enhancing competitiveness, enabling development, and bringing progress to all levels of society.

The career opportunities for computer science and information technology graduates are plenty and growing. Programming and software development, Data Scientists, Data Analysts, information systems operation and management, telecommunications and networking, computer science research, web and Internet, graphics and multimedia, training and support, and computer industry specialists are some of the opportunities the graduates find. The School of Computing and Information Technology at REVA UNIVERSITY offers B. Tech., Computer Science and Information Technology, an undergraduate program to create motivated, innovative, creative and thinking graduates to fill ICT positions across sectors who can conceptualize, design, analyze, and develop ICT applications to meet the modern day requirements. The B. Tech. in Computer Science and Information Technology curriculum developed by the faculty at the School of Computing and Information Technology is outcome based and it comprises required theoretical concepts and practical skills in the domain. By undergoing this program, students develop critical, innovative, creative thinking and problem solving abilities for a smooth transition from academic to real-life work environment. In addition, students are trained in interdisciplinary topics and attitudinal skills to enhance their scope. The above mentioned features of the program, advanced teaching and learning resources and experience of the faculty members with their strong connections with ICT sector makes this program unique.

Program Educational Objectives (PEO's)

After few years of graduation, the graduates of B. Tech CS&IT will:

PEO-1: Pursue higher education in the core or allied areas of Computer Science and Information Technology.

PEO-2: Have technical career in the core or allied areas of Computer Science and Information Technology or start entrepreneurial activity for the growth of the economy.

PEO-3: Continue to learn and to adapt to ever changing technologies in the core or allied areas of Computer Science and Information Technology.

Program Outcomes (PO's)

On successful completion of the program, the graduates of B. Tech CS & IT program will be able to:

PO-1: Engineering knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO-2: Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO-3: Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO-4: Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO-5: Engineering tool usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO-6: The engineer and The world: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO-7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).

PO-8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams

PO-9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO-10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO-11: Life-Long Learning: Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Knowledge and Attitude Profile (WK):

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Specific Outcomes (PSO's)

On successful completion of the program, the graduates of B. Tech CS IT program will be able to:

PSO-1: Apply the knowledge of mathematics, Computer Science and Information Technology to solve complex problems in CS and IT.

PSO-2: Analyze, design, develop solutions and conduct investigations in the domains of database, networks and security, system software and system administration.

PSO-3: Apply appropriate techniques, use modern programming languages, tools, and packages for quality software development of CSIT.

ACADEMIC REGULATIONS

B. Tech CSIT, Degree Programs (Applicable for the programs offered for 2025-26 Batch)

(Framed as per the provisions under Section 35 (ii), Section 7 (x) and Section 8 (xvi) & (xxi) of the REVA University Act, 2012)

THESE ACADEMIC REGULATIONS ARE UNDER CHOICE BASED CREDIT SYSTEM AND CONTINUOUS ASSESSMENT GRADING PATTERN (CBCS-CAGP)

1. Title and Commencement:

1.1 These Regulations shall be called **"REVA University Academic Regulations – B. Tech., Degree Program for the batch of students admitted for AY 2025-26 subject to amendments from time to time by the Academic Council on recommendation of respective Board of Studies and approval of Board of Management"**

1.2 These Regulations shall come into force from the date of assent of the Chancellor.

2. The Programs:

These regulations cover the following B. Tech., Degree programs of REVA University offered for the admitted batch during AY 2025-26 under respective schools.

SL No.	Name of the School	Name of the Program
1	School of Civil Engineering	B Tech in Civil Engineering
2	School of Computing and Information Technology	B Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning)
		B Tech in Computer Science and Information Technology
		B Tech in Information Science and Engineering
3	School of Computer Science and Engineering	B Tech in Computer Science and Engineering
		B Tech in Artificial Intelligence and Data Science
		B Tech in Computer Science and Engineering (Internet of Things and Cyber Security including Blockchain Technology)
4	School of Electrical and Electronics Engineering	B Tech in Electrical and Electronics Engineering
5	School of Electronics and Communication Engineering	B Tech in Electronics and Communication Engineering
		B Tech in Electronics and Computer Engineering
		B Tech in Robotics and Artificial Intelligence
6	School of Mechanical Engineering	B Tech in Mechanical Engineering
		B Tech in Mechatronics Engineering
		B.Tech in Aerospace Engineering
7	Department of Agricultural Engg.	B Tech in Agricultural Engineering

3. Duration and Medium of Instructions:

3.1 Duration: The duration of the B Tech degree program shall be **FOUR** years comprising of **EIGHT** Semesters. A candidate can avail a maximum of 16 semesters - 8 years as per double duration norm, in one stretch to complete B. Tech degree, including blank semesters, if any. Whenever a candidate opts for blank semester, he/she has to study the prevailing courses offered by the School when he/she resumes his/her studies. In case of lateral Entry batch, maximum 12 semesters or 6 years as per double duration norm.

3.2 The medium of instruction shall be English.

4. Definitions:

4.1 Course: "Course" means a subject, either theory or practical or both and project, listed under a program; Example: "Fluid Mechanics" in B. Tech Civil Engineering program, "Signals and Systems" in B. Tech., Electronics and Communication Engg are the examples of courses to be studied under respective programs.

Every course offered will have three components associated with the teaching-learning process of the course, namely, L, T and P, where,

L stands for **Lecture** session consisting of classroom instruction.

T stands for **Tutorial** session, an Interactive session including discussions, self-study, desk work, seminar presentations, and other methods to deepen understanding.

P stands for **Practice** session and it consists of hands-on experience such as laboratory experiments, field studies, case studies, project based learning or course end projects and self-study courses that equip students to acquire the required skill component.

4.2 Classification of Courses

Courses offered are classified as follows:

4.2.1 Foundation Course (FC): The foundation Course is basic course which should be completed successfully as a part of graduate degree program irrespective of the program of study.

4.2.2 Professional Core Course (also known as Hard Core(HC) Course): The **Professional Core** is a core course in the main branch of study and related branch(es) of study, if any, that the candidates have to complete compulsorily.

4.2.3 Professional Elective Course (also known as Soft Core (SC) Course): Professional Elective course is a choice or an option for the candidate to choose a course from a pool of courses from the main branch of study or from a sister/related branch of study which supports the main branch of study.

4.2.4 Open Elective Course (OE): An elective course chosen generally from other discipline / subject, with an intention to seek exposure to the basics of subjects other than the main discipline the student is studying is called an **Open Elective Course.**

4.2.5 Audit Course (also known as Non-Credit Course /Mandatory Course(MC)): These courses are mandatory for students joining B.Tech. Program and students have to successfully complete these courses before the completion of degree.

4.2.6 Project Work / Dissertation: Project work / Dissertation work is a special course involving application of knowledge in solving / analyzing /exploring a real life situation / difficult problems to solve a multivariable or complex engineering problems. The project will be conducted in two phases, phase-I, consists of literature survey, problem identification, formulation and methodology. In Phase-II, student should complete the project work by designing or creating an innovative process or development of product as an outcome. A project work is carried out as minor project in 3rd year and major project in 4th year with appropriate credits allocated.

4.2.7 Skill Development Course (SDC): It is a practice based course introduced in first year, second year and third year that lead to advanced job skills as per current industry/societal requirements to enhance high employability index of graduates. It may also lead to a certificate, diploma and advanced diploma, etc.

4.2.8 Emerging Technology Course (ETC): This course is introduced in the first year, focusing on the latest advancements and innovations in technology. It aims to equip students with cutting-edge knowledge and skills relevant to emerging fields.

4.3 "Program" means the academic program leading to a Degree, Post Graduate Degree, Post Graduate Diploma Degree or such other degrees instituted and introduced in REVA University.

5. Eligibility for Admission:

5.1. The eligibility criteria for admission to B Tech Program of 4 years (8 Semesters) is given below:

Sl. No.	Program	Duration	Eligibility
1	Bachelor of Technology (B. Tech)	4 Years (8 Semesters)	A. FOR GENERAL MERIT CANDIDATES: A candidate who has Passed in 2nd PUC / 12 Std / Equivalent Examination with English as one of the Language and obtained a Minimum of 45% of Marks in aggregate in Physics and Mathematics as compulsory subjects along with Chemistry / Bio Technology / Biology / Computer Science / Electronics as optional subjects in the qualifying examination is eligible to pursue in under graduate programs (BE). B. FOR SC/ST & OBC (Cat-I, 2A, 2B, 3A 3B) CATEGORY CANDIDATES: A candidate who has Passed in 2nd PUC / 12 Std / Equivalent Examination with English as one of the Language and obtained a Minimum of 40% of Marks in aggregate in Physics and Mathematics as compulsory

Sl. No.	Program	Duration	Eligibility
			subjects along with Chemistry / Bio Technology / Biology / Computer Science / Electronics as optional subjects in the qualifying examination is eligible to pursue in under graduate pogroms (BE). C. The marks obtained by the candidate in Biotechnology/Biology/Computer Science / Electronics in the qualifying examination will be considered in the place of Chemistry in case the marks obtained in Chemistry is less for the required aggregate percentage for the pursue of determination of eligibility.
2	Bachelor of Technology (B Tech-Lateral Entry program)	3 Years (6 Semesters)	A. candidate who has passed Qualifying Examination i.e. any diploma examination or equivalent examination and obtained an aggregate minimum of 45 % (for General Merit Candidates) of marks taken together in all the subjects of the final year (i.e.,fifth and six semesters) diploma examination (Q E) is eligible for admission to B E-Courses and 40% of marks in Q. E. in case of SC, ST and Backward Classes of Karnataka Candidates. Passed B.Sc Degree from a recognized University as defined by UGC, with at least 45% marks (40% in case of candidates belonging to SC/ST category) and passed XII standard with mathematics as a subject. B. Provided that in case of students belonging to B.Sc. Stream, shall clear the subjects of Engineering Graphics / Engineering Drawing and Engineering Mechanics of the first year Engineering program along with the second year subjects. C. Provided further that, the students belonging to B. Sc. Stream shall be considered only after filling the seats in this category with students belonging to the Diploma stream. D. Provided further that student, who have passed Diploma in Engineering & Technology from an AICTE approved Institution or B. Sc., Degree from a recognized University as defined by UGC, shall also be eligible for admission to the first year Engineering Degree courses subject to vacancies in the first year class in case the vacancies at lateral entry are exhausted. However the admissions shall be based strictly on the eligibility criteria as mentioned in A, B, D, and E above.

Sl. No.	Program	Duration	Eligibility
			E. Passed D.Voc. Stream in the same or allied sector. (The Universities will offer suitable bridge courses such as Mathematics, Physics, Engineering drawing, etc., for the students coming from diverse backgrounds to achieve desired learning outcomes of the program)
3	Bachelor of Technology (B Tech)		Any candidate with genuine reason from any University / Institution in the country upon credit transfer could be considered for lateral admission to the respective semester in the concerned branch of study, provided he/she fulfils the University requirements.

5.2 Provided further that the eligibility criteria are subject to revision by the Government Statutory Bodies, such as AICTE, UGC from time to time.

6. Courses of Study and Credits

6.1 Each course of study is assigned with certain credit value

6.2 Each semester is for a total duration of 20 weeks out of which 16 weeks dedicated for teaching and learning, CIA and the remaining 4 weeks for SEE, evaluation and result declaration.

6.3 The credit hours defined as below:
In terms of credits, L refers to lecture hour (theory) credit per week, that indicate every one hour lecture per week of L amounts to 1 credit per Semester; T and P refer to tutorial hours and practice hours credit per week, that indicate every two hours of T and P per week amounts to 1 credit per Semester or a three hour session of T / P amounts to 2 credits per semester.

The total duration of a semester is 20 weeks inclusive of semester-end examination.
The following table describes credit pattern

a. The concerned BoS will choose the convenient Credit Pattern for every course based on size and nature of the course.

Table -2: Credit Pattern

Lectures (L)	Tutorials (T)	Practice (P)	Credits (L:T:P)	Total Credits	Total Contact Hours
4	2	0	4:1:0	5	6
3	2	0	3:1:0	4	5
3	0	2	3:0:1	4	5
2	2	2	2:1:1	4	6
0	0	6	0:0:3	3	6
4	0	0	4:0:0	4	4
2	0	0	2:0:0	2	2
0	0	2	0:0:1	1	2

7. Different Courses of Study:

Different **Courses of Study** are labeled as follows:

a. Foundation Course (FC)

b. Professional Core Course (Hard Core(HC))

c. Professional Elective Course (Soft Core(SC))

d. Open Elective Course (OE)

e. Skill Development Course (SDC)

f. Audit Course (Non-credit Course/ Mandatory Course) (MC)

g. Project Work / Dissertation: A project work is carried out as minor project in 3rd year and major project in 4th year with appropriate credits allocated. These are defined under Section 4.2.6 of this regulation.

h. Emerging Technology Course(ETC)

8. Credits and Credit Distribution

A candidate must earn 160 credits for successful completion of B Tech degree with the distribution of credits for different courses with the credit distribution given in the scheme of study.

8.1 The concerned BOS based on the credits distribution shall prescribe the credits to various types of courses listed in section 4.2 and shall assign title to every course thereon.

8.2 Every course including project work, practical work, field work, self-study elective should be entitled as per the list declared in section 4.2. However, as per AICTE, the credit distribution for various category of courses is given below in the table.

Sl. No.	Course Category	Abbreviation (AICTE)	Suggested breakup of credits (AICTE)	Credit breakup (REVA)
1	Humanities and Social Sciences including Management courses (HSMC)	HSMC	12	8
2	Basic Science Courses	BSC	25	18
3	Engineering Science courses including workshop, drawing, basics of electrical/mechanical/computer etc	ESC	24	18
4	Program Core Courses	PCC	48	63
5	Program Elective courses relevant to chosen specialization/branch	PEC	18	18
6	Open subjects – Electives from other technical and /or emerging subjects	OE	18	6
7	Project work, seminar and internship in industry or elsewhere	PROJ	15	23
8	Audit Courses (Mandatory Course)	MC	-	-
9	Skill Development Courses (SDC)/ETC/AEC	ETC/AEC/SDC		6
			160	160

8.3 The concerned BOS shall specify the desired Program Educational Objectives, Program Outcomes, Program Specific Outcomes and Course Outcomes while preparing the curriculum of a particular program. A candidate can enroll for a maximum of 26 credits and a minimum of 16 credits per Semester. However, he / she may not successfully earn a maximum of 26 credits per semester. This maximum of 26 credits does not include the credits of courses carried forward by a candidate.

8.4 **Only such full-time candidates who register for a minimum prescribed number of credits in each semester from I semester to VIII semester and complete successfully 160 credits in 8 successive semesters shall be considered for declaration of Ranks, Medals, Prizes** and are eligible to apply for Student Fellowship, Scholarship, Free ships, and such other rewards / advantages which could be applicable for all full time students.

8.5 **Minor degree/ Honor Degree:**

To acquire Add on Proficiency Diploma/ Minor degree/ Honor Degree: a candidate can opt to complete a minimum of 18 extra credits either in the same discipline /subject or in different discipline / subject in excess to 160 credits for the B Tech Degree program. The Add on Proficiency Certification / Diploma/ Minor degree/ Honor Degree: so issued to the candidate contains the courses studied and grades earned.

9 Assessment and Evaluation

9.1 The Scheme of Assessment will have two parts, namely;

i. Continuous Internal Assessment (CIA); and
ii. Semester End Examination (SEE)

9.2 Assessment and Evaluation of each Course with ≥ 2 credits shall be for 100 marks. The CIA and SEE of UG Engineering programs shall carry 50 marks each (i.e., 50 marks for CIA and 50 for marks SEE (detailed break up is as indicated in section 9.27).

For Theory courses with < 2 credits, assessment and evaluation shall be for 50 marks. The CIA and SEE shall carry 25 marks each (i.e., 25 marks for CIA and 25 marks for SEE). (Detailed break up indicated in section 9.28).

9.3 The 50 marks of CIA shall comprise of:

Internal Assessment Test (IA Test)	40 marks
Assignments / Seminars / Model Making / Integrated Lab / Project Based Learning / Quizzes, etc.	10 marks

9.4 There shall be **two Internal Assessment Tests** (IA tests) are conducted as per the schedule announced below. **The Students shall attend both the Tests compulsorily.**

- 1st test is conducted for 20 marks during **8th week** of the Semester;
- 2nd test is conducted for 20 marks during **15th week** of the of the Semester;

9.5 The coverage of syllabus for the said tests shall be as under:

- The question paper of the **1st test should be based on first 50% of the total syllabus**.

- The question paper of the 2^{nd} **test should be based on remaining 50 % of the total syllabus.**
- An assignments must be designed to cover the entire syllabus.

9.6 There shall be two Assignments / Project Based Learning / Field Visit / Quiz test carrying 10 marks covering the entire syllabus.

9.7 SEE for 50 marks practical exam shall be held during 16^{th} and 17^{th} week of the semester.

9.8 SEE for 50 marks theory exam shall be held during 18^{th} 19^{th} and 20^{th} week of the semester and it should cover entire syllabus.

9.9 IA test paper is set for a maximum of 40 marks to be answered in 1.5 hours duration (for 1 credit course, exam is conducted for 25 marks with a duration of 1 hour). A test paper can have 5 main questions. Each main question is set for 10 marks. The main question can have 2-3 sub questions all totaling 10 marks. Students are required to answer any 4 main questions. Each question is set using Bloom's action verbs. The questions must be set to assess the course outcomes described in the course document with the choice is given in questions.

9.10 The question papers for IAs shall be set by the internal faculty who have taught the course. If the course is taught by more than one faculty all the them together shall devise a common question paper(s). However, these question papers shall be scrutinized by the Question Paper Scrutiny Committee (internal BoE members) to bring the quality and uniformity in the question paper.

9.11 The evaluation of the answer scripts shall be done by the internal faculty who have taught the course and set the test paper. After evaluation of answer booklets faculty must distribute to students. If any corrections are noticed by students, faculty have to justify the award of marks. The final marks are to be declared only after such corrections. school/Department has to take signature for the marks obtained.

9.12 Assignment/seminar/Project based learning/simulation based problem solving/field work should be set in such a way, students be able to apply the concepts learnt to a real life situation and students should be able to do some amount of self-study and creative thinking. While setting assignment care should be taken such that the students will not be able to plagiarize the answer any other resources. Course faculty at his/her discretion can design the questions as a small group exercise or individual exercise. This should encourage collaborative learning and team learning and self-study.

9.13 CIA marks must be decided well before the commencement of SEE.

9.14 The SEE theory question paper is designed to comprehensively evaluate student's understanding and mastery of the course content. A total of 8 main questions will be crafted to cover the entire syllabus, out of which students are required to answer any 5 questions in full. Each main question will carry a maximum of 20 marks and may include 3 to 4 sub-questions. The maximum marks for each course is 100, and the duration is 3 hours. All questions must be formulated using Bloom's action verbs to ensure alignment with cognitive learning objectives. Additionally, the questions must be meticulously crafted to assess the student outcomes detailed in the course document, ensuring thorough coverage of all the course outcomes.

9.15 There shall be minimum three sets of question papers for the SEE, of which one set along with scheme and solution of examination shall be set by the external examiners and two sets along with scheme of examination shall be set by the internal examiners. All the question papers shall be scrutinized by the Board of Examiners (BoE). It shall be responsibility of the BOE particularly Chairman of the BOE to maintain the quality and standard of the question papers and as well the coverage of the entire syllabus of the course.

9.16 There shall be single evaluation by the examiners for each paper. However, there shall be moderation by one of the senior examiners, either internal or external.

9.17 Board of Examiners, question paper setters and any member of the staff connected with the examination are required to maintain integrity of the examination system and the quality of the question papers.

9.18 There shall also be a **Program Assessment Committee (PAC)** comprising at-least 3 faculty members having subject expertise who shall after completion of examination process and declaration of results review the results sheets, assess the performance level of the students, measure the attainment of course outcomes, program outcomes and assess whether the program educational objectives are achieved and report to the Director of the School.

9.19 The report provided by the PAC shall be the input to the Board of Studies to review and revise the scheme of instruction and curriculum of respective program.

9.20 During unforeseen situation like the Covid-19, the tests and examination schedules, pattern of question papers and weightage distribution may be designed as per the convenience and suggestions of the board of examiners in consultation with Controller of Examination and Vice chancellor.

9.21 University may decide to use available modern technologies for writing the IAs and SEE by the students instead of traditional pen and paper.

9.22 Any deviations required to the above guidelines can be made with the written consent of the Vice Chancellor.

9.23 Online courses may be offered as per UGC norms.
For online course assessment guidelines would be as follows:

a. If the assessment is done by the course provider, then the school can accept the marks awarded by the course provider and assign the grade as per REVA University norms.

b. If the assessment is not done by the course provider, then the assessment is organized by the concerned school and the procedure explained in the regulation will apply.

c. In case a student fails in an online course, s/he may be allowed to repeat the course and earn the required credits.

IAs for online courses could be avoided and will remain at the discretion of the school.

9.24 The online platforms identified could be SWAYAM, NPTEL, Coursera, Edx.org, Udemy, Udacity and any other internationally recognized platforms like MIT online, Harvard online etc.

9.25 Mapping of one or two credit online courses would be:

4 week online course – 1 credit

8 week online course / MOOC – 2 credits

12 week online course / MOOC – 3 credits

9.26 **Summary of Internal Assessment, Semester End Examination and Evaluation** Schedule is provided in the table given below (for theory courses having Credits >=2).

Summary of Internal Assessment and Evaluation Schedule

Sl. No.	Type of Assessment	When	Syllabus Covered	Max Marks	Scaled down to	Date by which the process must be completed
1	Test-1	During 8th week	First 50%	40	20	9th week
2	Test -2	During 15th Week	Remaining 50%	40	20	16th Week
3	Assignment 1/ Quiz - 1	Every week till Test-1	First 50%	10	05	9th Week
4	Assignment 2 / Quiz - 2	Every week during Test-1 and Test-2	Remaining 50%	10	05	16th Week
5	SEE	18th to 20th Week	100%	100	50	20th Week

*Minimum of 30 marks to be scored out of 100 marks in SEE (i.e., Minimum of 15 marks out of 50 marks in SEE after scaling down). IA (50 marks) and SEE (50 marks) are added for Semester End result. Minimum of 40 marks to be scored out of 100 for result to be announced as PASS.

9.27 Summary of Internal Assessment, Semester End Examination and Evaluation Schedule is provided in the table given below (for theory courses having Credit 1).

Summary of Internal Assessment and Evaluation Schedule

Sl. No.	Type of Assessment	When	Syllabus Covered	Max Marks	Reduced to	Date by which the process must be completed
1	Test-1/Assignment/Quiz	During 8th week	First 50%	25	12.5	8th week
2	Test-2/Assignment/Quiz	During 15th Week	Remaining 50%	25	12.5	15th Week
5	SEE	18th to 20th Week	100%	50	25	20th Week

Minimum of 15 marks to be scored out of 50 in SEE (i.e., Minimum of 8 marks out of 25 in SEE after scaling down). IA (25marks) and SEE (25 marks) are added for Semester End result. Minimum of 20 marks to be scored out of 50 for result to be announced as PASS.

10 Assessment of Students Performance in Practical Courses

Lab courses are of two types: integrated labs and separate labs.

The performance in the practice tasks / experiments shall be assessed on the basis of:

a. Knowledge of relevant processes;
b. Skills and operations involved;
c. Results / products including calculation and reporting

10.1 Assessment of lab courses

10.1.1 Assessment of Separate lab course

The 50 marks of CIA is based on the performance of students in each lab experiment for a lab course that shall further be allocated as under:

i	Conduction of regular practical / experiments throughout the semester(Continuous evaluation)	20 marks
ii	Maintenance of lab records	10 marks
iii	Performance of internal lab test to be conducted after completion of all the experiments before last working day of the semester	20 marks
	Total	**50 marks**

10.2 Assessment of integrated lab course

The 10 marks meant for CIA is based on the performance of students in each lab experiment for integrated lab course that shall further be allocated as under.

Integrated lab is evaluated and awarded marks should meet the requirement of assignment/quiz/field work component of respective theory course having integrated lab component. No separate assignment/quiz/filed work is assessed for such courses.

i	Conduction of regular practical / experiments throughout the semester (continuous evaluation)	05 marks
ii	Maintenance of lab records and performance of internal lab test to be conducted after completion of all the experiments before last working day of the semester	05 marks
	Total	**10 marks**

10.3 The 50 marks meant for SEE in case of separate lab course shall be allocated as under:

i	Conduction of practical (experiment)	30 marks
ii	Write up about the experiment/tabulation/results/inference	10 marks
iii	Viva Voce	10 marks
	Total	**50 marks**

Note: No Separate SEE for integrated lab course

10.4 The duration for semester-end practical examination shall be decided by the concerned School Board. Semester End Practical Examination shall be conducted and assessed by appointing external faculty members of reputed institutions/Universities with subject expertise as external examiners, along with faculty members from respective schools as internal examiners.

10.5 For MOOC and Online Courses assessment shall be decided by the BOS of the School.

For >= 2 credit courses

i	IA-I	25 marks
ii	IA-2	25 marks
iii	Semester end examination by the concern school board (demo, test, viva voice etc.)	50 marks
	Total	**100 marks**

For 1 credit courses

i	IA (Performance of internal test to be conducted after completion of entire syllabus)	25 marks
iii	Semester end examination by the concern school board (demo, test, viva voice etc.)	25 marks
	Total	**50 marks**

11. Evaluation of Minor Project / Major Project / Dissertation >4 credit:

Right from the initial stage of defining the problem, the candidate must submit the progress reports periodically and present his/her progress in the form of seminars in addition to the regular discussion with the supervisor. At the end of the semester, the candidate must submit final report of the project / dissertation for final evaluation. The components of evaluation are as follows:

Component – I	Periodic Progress and Progress Reports (25%)
Component – II	Demonstration and Presentation of work (25%)
Component – III	Evaluation of Report (50%)

i	Phase-1 presentation	50 marks
ii	Phase-II presentation	50 marks
III	SEE	100 marks
	Total	**200 marks**

12. Evaluation of mandatory courses: Students should maintain minimum of 75% attendance to appear for SEE of Mandatory course. The SEE shall be conducted using Multiple Choice Questions (MCQ)/Descriptive/Presentation/ Activity based/ Viva-voce type of assessment, which will be at the discretion of course teachers and approval from Director of respective schools offering the Mandatory Course.

The performance of the mandatory course is evaluated through SEE component only. However, course teachers can conduct class tests during IA2. The minimum 40% of SEE score is mandatory for students to PASS.

13. Evaluation of **Skill Development Courses:** The concerned BoS shall recommend to conduct test/demo/viva-voce/MCQ to test the student knowledge.

14. Requirements to Pass a Course:

A candidate's performance from CIA and SEE will be in terms of scores, and the sum of CIA and SEE scores will be for a maximum of 100 marks (CIA = 50 , SEE = 50) and have to secure a minimum of 40% to declare pass in the course. However, a candidate must secure a minimum of 30% (15 marks) out of 50 marks in SEE, which is compulsory.

The Grade and the Grade Point: The Grade and the Grade Point earned by the candidate in the subject will be as given below:

SGPA/CGPA	Grade Point	Letter Grade	Performance	FGP Qualitative Index
9>=CGPA<=10	10	O	Outstanding	First Class with Distinction
8>=CGPA<=9	9	A+	Excellent	
7>=CGPA<=8	8	A	Very Good	First Class
6>=CGPA<=7	7	B+	Good	
5.5>=CGPA<=6	6	B	Above Average	Second Class
5>=CGPA<=5.5	5.5	C+	Average	
4>=CGPA<=5	5	C	Satisfactory	Pass
<4CGPA	0	F	Not Satisfactory	Unsuccessful

a. Computation of SGPA and CGPA

The Following procedure to compute the Semester Grade Point Average (SGPA).

The SGPA is the ratio of sum of the product of the number of credits with the grade points scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student in a given semester, i.e : **SGPA (Si) = ∑(Ci x Gi) / ∑Ci** where Ci is the number of credits of the ith course and Gi is the grade point scored by the student in the ith course.

Illustration for Computation of SGPA and CGPA

Illustration No. 1

Course	Credit	Grade Letter	Grade Point	Credit Point (Credit x Grade)
Course 1	3	A+	9	3X9=27
Course 2	3	A	8	3X8=24
Course 3	3	B+	7	3X7=21
Course 4	4	O	10	4X10=40
Course 5	1	C	5	1X5=5
Course 6	2	B	6	2X6=12
Course 7	3	O	10	3X10=30
	19			159

Thus, **SGPA = 159 ÷ 19 = 8.37**

Illustration No. 2

Course	Credit	Grade letter	Grade Point	Credit Point (Credit x Grade point)
Course 1	4	A	8	4X8=32
Course 2	4	B+	7	4X7=28
Course 3	3	A+	9	3X9=27
Course 4	3	B+	7	3X7=21
Course 5	3	B	6	3X6=18
Course 6	3	C	5	3X5=15
Course 7	2	B+	7	2X7=21
Course 8	2	O	10	2X10=20
	24			182

Thus, SGPA = 182 ÷ 24 = 7.58

Illustration No.3

Course	**Credit**	**Grade Letter**	**Grade Point**	**Credit Point (Credit x Grade point)**
Course 1	4	O	10	4 x 10 = 40
Course 2	4	A+	9	4 x 9 = 36
Course 3	3	B+	7	3 x 7 = 21
Course 4	3	B	6	3 x 6 = 18
Course 5	3	A+	9	3 x 9 = 27
Course 6	3	B+	7	3 x 7 = 21
Course 7	2	A+	9	2 x 9 = 18
Course 8	2	A+	9	2 x 9 = 18
	24			**199**

Thus**, SGPA = 199 ÷ 24 = 8.29**

b. Cumulative Grade Point Average (CGPA):

Overall Cumulative Grade Point Average (CGPA) of a candidate after successful completion of the required number of credits (168) for B. Tech degree in Engineering & Technology is calculated taking into account all the courses undergone by a student over all the semesters of a program, i. e : **CGPA = ∑(Ci x Si) / ∑Ci**

Where Si is the SGPA of the ith semester and Ci is the total number of credits in that semester.

Illustration:

CGPA after Final Semester

Semester (ith)	No. of Credits (Ci)	SGPA (Si)	Credits x SGPA (Ci X Si)
1	21	6.83	21 x 6.83 = 143.43
2	20	7.29	20 x 7.29 = 145.8
3	25	8.11	25 x 8.11 = 202.75
4	25	7.40	25 x 7.40 = 185
5	22	8.29	22 x 8.29 = 182.38
6	23	8.58	23 x 8.58 = 197.34
7	14	9.12	14 x 9.12 = 127.68
8	10	9.25	10 x 9.25 =92.50
Cumulative	**160**		**1276.88**

Thus,

$$CGPA = \frac{1276.88}{160} = 7.98$$

c. Conversion of grades into percentage:

Conversion formula for the conversion of CGPA into Percentage is:

Percentage of marks scored = CGPA Earned x 10

Illustration: CGPA Earned 8.02 x 10=80.2

d. The SGPA and CGPA shall be rounded off to 2 decimal points and reported in the transcripts.

Classification of Results

The final grade point (FGP) to be awarded to the student is based on CGPA secured by the candidate and is given as follows.

CGPA	Grade (Numerical Index) G	Letter Grade	Performance	FGP Qualitative Index
9 >= CGPA 10	10	O	Outstanding	Distinction
8 >= CGPA < 9	9	A+	Excellent	
7 >= CGPA < 8	8	A	Very Good	First Class
6 >= CGPA < 7	7	B+	Good	
5.5> = CGPA < 6	6	B	Above average	Second Class
> 5 CGPA < 5.5	5.5	C+	Average	
> 4 CGPA <5	5	C	Satisfactory	Pass
< 4 CGPA	0	F	Unsatisfactory	Unsuccessful

Overall percentage=10*CGPA

e. **Provisional Grade Card**: The tentative / provisional grade card will be issued by the Controller of Examinations at the end of every semester indicating the courses completed successfully. The provisional grade card provides **Semester Grade Point Average (SGPA).**

f. **Final Grade Card:** Upon successful completion of B Tech Degree a Final Grade card consisting of grades of all courses successfully completed by the candidate will be issued by the Controller of Examinations.

14.1 Attendance Requirement

14.1.1. All students must attend every lecture, tutorial and practical classes.

14.1.2. Student must maintain a minimum attendance of 75% in each course (Theory and Practical) and 75% attendance in aggregate of all courses in a semester, with a provision of condonation of 10% of the attendance by the Vice-Chancellor on the specific recommendation of the Director of the School.

14.1.3. In case a student is on approved leave of absence (e g:- representing the University in sports, games or athletics, placement activities, NCC, NSS activities and such others) and / or any other such contingencies like medical emergencies, the attendance requirement shall be minimum of 65% of the classes taught.

14.1.4. Any student with less than 75% of attendance in individual courses of respective semester including practical courses / field visits etc., shall not be permitted to appear to SEE in the respective course.

15. Re-Registration / Re-Admission

15.1 In case a candidate's class attendance in aggregate of all courses in a semester is less than 75% or as stipulated by the University, such a candidate is considered as dropped the semester and is not allowed to appear for semester end examination and he / she shall have to seek re-admission to that semester during subsequent semester / year within a stipulated period.

15.2 In such case where in a candidate drops all the courses in a semester due to personal reasons, it is considered that the candidate has dropped the semester and he / she shall seek re-admission to such dropped semester.

16. Absence during Internal Test:

In case a student has been absent from an internal test due to the illness or other contingencies he / she may give a request along with necessary supporting documents and certification from the concerned class teacher / authorized personnel to the concerned Director of the School, for conducting a separate internal test. The Director of the School may consider such a request depending on the merit of the case and after consultation with course instructor and class teacher, and arrange to conduct a special internal test for such candidate(s) well in advance before the Semester End Examination of that respective semester. Under no circumstances internal tests shall be held / assignments are accepted after Semester End Examination.

17. Provision for Appeal

If a candidate is not satisfied with the evaluation of Internal Assessment components (Internal Tests and Assignments), he/she can approach the Grievance Cell with the written submission together with all facts, the assignments, and test papers, which were evaluated. He/she can do so before the commencement of respective semester-end examination. The Grievance Cell is empowered to revise the marks if the case is genuine and is also empowered to levy penalty as prescribed by the University on the candidate if his/her submission is found to be baseless and unduly motivated. This Cell may recommend for taking disciplinary/corrective action on an evaluator if he/she is found guilty. The decision taken by the Grievance committee is final.

17.1 Grievance Committee:

In case of students having any grievances regarding the conduct of examination, evaluation and announcement of results, such students can approach Grievance Committee for redressal of grievances.

For every program there will be one grievance committee. The composition of the grievance committee is as follows:-

- The Controller of Examinations - Ex-officio Chairman / Convener
- One Senior Faculty Member (other than those concerned with the evaluation of the course concerned) drawn from the school / department/discipline and/or from the sister schools / departments/sister disciplines – Member.
- One Senior Faculty Members / Subject Experts drawn from outside the University school / department – Member.

18. Eligibility to Appear for Semester End Examination (SEE)

Only those students who fulfil a minimum of 75% attendance in aggregate of all the courses including practical courses / field visits etc., and 70% attendance in each courses shall be eligible to appear for Semester End Examination

19. Provision for Supplementary Examination

In case a candidate fails to secure a minimum of 30% marks out of 50 (15 marks) in Semester End Examination (SEE) and a minimum of 40% marks (out of 100) together with IA and SEE to declare pass in the course, such candidate shall seek supplementary examination of only such course(s) wherein his / her performance is declared unsuccessful. The supplementary examinations are conducted after the announcement of even/Odd semester examination results. The candidate who is unsuccessful in each course(s) shall appear for supplementary examination of odd and even semester course(s) to seek for improvement of the performance.

a. A student who failed in any course is eligible to take supplementary exam under following category for each course: either to improve internal marks (IA1, IA2, and Assignment/Quiz), or to improve SEE.

b. Supplementary exam is permitted only during summer vacation (between even and odd semester break)

c. Eligibility to register for supplementary exam is that the student should have maintained pre-requisite attendance of ≥ 75% in respective semester.

d. No separate additional classes would be conducted for the students availing this facility.

e. Every student should pay the supplementary exam fee for each course as prescribed by the university.

20. Provision to Carry Forward the Failed Subjects / Courses:

Students who have failed in courses totaling 16 credits or fewer across both odd and even semesters combined shall be eligible to proceed to the next academic year(s) of their B.Tech program including to lateral entry students. For vertical progression, students must successfully complete all the first year courses (from both semesters) to qualify for admission into the Fourth Year of the B.Tech Program.

Case 1: A student who has failed in courses with a maximum of 16 credits in I and II semester combined, shall be permitted to move to the 3rd semester in the succeeding academic year.

Case 2: A student who has failed in courses totaling a maximum of 16 credits across the I to IV semesters combined, shall be permitted to move to the V semester in the succeeding academic year.

Case 3: A student who has failed in courses totaling a maximum of 16 credits across the III to VI Semesters combined shall be permitted to seek admission to the 4th year VII Semester in the succeeding academic year, provided that all courses of the I and II semesters have been successfully completed.

21. Re-evaluation of Answer Scripts and Announcement of Re-evaluation Results

After declaration of the results of programs within next 10 days, if any candidate wishes to apply for Photocopy/Revaluation (only theory courses), s/he shall apply to the Controller of Examinations, by paying the prescribed fees notified by the University from time to time. The photocopies of the said answer books shall be made available within next TEN working days after the last date prescribed for receipt of the application at the Office of the Controller of Examinations. Photocopies will not be issued for practical/drawing/audit courses.

22. Results of Re-Evaluation will be announced within TWENTY working days (except for third evaluation).

23. Regarding any specific case of ambiguity and unsolved problem, the decision of the Vice- Chancellor shall be final.

24. All assessments must be done by the respective Schools as per the guidelines issued by the Controller of Examinations. However, the responsibility of announcing final examination results and issuing official transcripts to the students lies with the office of the Controller of Examinations.

25. For lateral entry students, the minimum credits to be earned for the award of the degree would be the credits earned in 3 years from 2nd year to 4th year.

SCHOOL OF COMPUTING AND INFROMATION TECHNOLOGY

Approved Scheme for First year B. Tech Engg.2025-26

I Semester												
S. No	Course Code	Title of the Course	HC/ FC /SC /OE /M C/E TC	Credit Pattern				Contact Hours/ Week	Examination			Course category (As per AICTE)
				L	T	P	Total Credit		CIE Marks	SEE Marks	Total Marks	
1	B24AS0103	Multivariable Calculus and Linear Algebra	FC	3	0	0	3	3	50	50	100	BSC
2	B24AS0106	Physics for Computer Science	FC	3	0	0	3	3	50	50	100	BSC
3	B24ME0105	Fundamentals of Mechanical Engineering	HC	3	0	0	3	3	50	50	100	ESC
4	B25CS0201	Business Analysis and Software Design	HC	2	0	0	2	2	50	50	100	ESC
5	B25CI0101	Introduction to C Programming	HC	1	1	0	2	3	50	50	100	ESC
6	B24EN0101	Internet of Things	HC	1	0	1	2	3	50	50	100	ESC
7	B25CI0102	Introduction to C Programming Lab	HC	0	0	1	1	2	25	25	50	ESC
8	B24AS0108	Physics for Computer Science Lab	HC	0	0	1	1	2	25	25	50	BSC
9	B24ME0102	Innovation and Entrepreneurship	FC	1	0	1	2	3	50	50	100	HSMC
10	B24AH0103	Communicative English	FC	0	0	1	1	2	25	25	50	HSMC
		TOTAL		**14**	**1**	**5**	**20**	**26**	**425**	**425**	**850**	
	TOTAL SEMESTER CREDITS		**20**									
	TOTAL CUMULATIVE CREDITS		**20**									
	TOTAL CONTACT HOURS		**26**									
	TOTAL MARKS		**850**									

II Semester												
S. No	Course Code	Title of the Course	HC/ FC /SC/ OE/ MC/ ETC	Credit Pattern				Contact Hours/ Week	Examination			Course category (As per AICTE)
				L	T	P	Total Credit	Contact hour	CIE Marks	SEE Marks	Total Marks	
1	B24AS0203	Probability and Statistics	FC	3	0	0	3	3	50	50	100	BSC
2	B25AS0105	Chemical Technology for Computing	FC	2	1	0	3	4	50	50	100	BSC
3	B25CI0201	Advanced C programming with Copilot	HC	1	1	0	2	3	50	50	100	ESC
4	B25EE0101	Electronics and Digital Logic	HC	3	0	0	3	3	50	50	100	ESC
5	B25CS0101	Python for Data Science	HC	2	0	0	2	2	50	50	100	ESC

6	B24ED0102	Fundamentals and Applications of Civil Engineering	HC	1	1	0	2	3	50	50	100	ESC
7	B25CI0202	Advanced C Programming with Copilot Lab	HC	0	0	1	1	2	25	25	50	PCC
8	B25EE0102	Electronics and Digital Logic Lab	HC	0	0	1	1	2	25	25	50	ESC
9	B25CS0102	Python for Data Science Lab	HC	0	0	1	1	2	25	25	50	ESC
10	B24EN0102	Finance and Management	FC	1	0	0	1	1	25	25	50	HSMC
11	B24CIET01	Engineering Exploration	HC	1	0	0	1	1	25	25	50	ESC
12	B25CSET01	AI Foundations for Engineers	ETC	1	0	0	1	1	25	25	50	ETC
13	B25CI0203	Grand Challenge (Extra credit 0-0-1)	MC									
		TOTAL		**15**	**3**	**3**	**21**	**27**	**450**	**450**	**900**	
	TOTAL SEMESTER CREDITS			**21**								
	TOTAL CUMULATIVE CREDITS			**41**								
	TOTAL CONTACT HOURS			**27**								
	TOTAL MARKS			**900**								

Detailed Syllabus
Semester 1

<table>
<tr><td>Course Title</td><td colspan="4">Multivariable Calculus and Linear Algebra</td><td colspan="2">Course Type</td><td colspan="2">HC</td></tr>
<tr><td>Course Code</td><td>B24AS0103</td><td>Credits</td><td colspan="2">3</td><td colspan="2">Class</td><td colspan="2">I Semester</td></tr>
<tr><td rowspan="5">Course Structure</td><td>LTP</td><td>Credits</td><td>Contact Hours</td><td>Work Load</td><td colspan="2" rowspan="2">Total Number of Classes Per Semester</td><td colspan="2" rowspan="2">Assessment in Weightage</td></tr>
<tr><td>Lecture</td><td>3</td><td>3</td><td>3</td></tr>
<tr><td>Tutorial</td><td>-</td><td>-</td><td>-</td><td rowspan="2">Theory</td><td rowspan="2">Practical</td><td rowspan="2">CIE</td><td rowspan="2">SEE</td></tr>
<tr><td>Practice</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>Total</td><td>3</td><td>3</td><td>3</td><td>42</td><td>-</td><td>50%</td><td>50%</td></tr>
</table>

COURSE OVERVIEW

This course introduces fundamental concepts of Vector Calculus, Differential Calculus, and Linear Algebra, which form the foundation for higher studies in mathematics, computer science, and engineering. The course covers essential vector operations such as velocity, acceleration, gradient, divergence, curl, and vector identities, along with scalar potentials and properties of solenoidal and irrotational fields. In Differential Calculus, students learn higher-order derivatives, Leibnitz theorem, Taylor’s and Maclaurin’s series expansions, L'Hospital’s rule for indeterminate forms, and applications of gradient descent methods. Linear Algebra topics include rank of matrices, echelon forms, system of linear equations, Gauss elimination, LU decomposition, linear transformations, eigenvalues and eigenvectors, diagonalization, and the Rayleigh power method.

The knowledge gained from this course is vital in computer science applications such as machine learning, optimization, data mining, computer graphics, scientific computing, cryptography, image processing, and network algorithms. By the end of the course, students will have a strong foundation in calculus and linear algebra for solving real-world computational and mathematical problems.

COURSE OBJECTIVE (S):

Students are able to learn

1. The fundamental concepts of vector calculus, series expansions, and L'Hospital’s rule.
2. The concepts of consistency and LU decomposition for solving linear systems.
3. The concepts of eigenvalue/eigenvector for various engineering problems.
4. The techniques like diagonalization and Rayleigh’s power method in real-life problems.

COURSE OUTCOMES(COs)

After the completion of the course, the student will be able to:

CO#	Course Outcomes	Pos	PSO
CO-1	Apply the concepts of gradient, divergence, curl, solenoidal and irrotational fields in vector calculus problems.	1,2,4	1
CO-2	Analyse the concepts of Taylor/Maclaurin and indeterminate forms	1,2,4	1
CO-3	Solve the homogeneous and non-homogeneous linear system equations converting in to matrix system.	1,2,4	1
CO-4	Solve Gauss elimination and LU decomposition methods to compute solutions efficiently.	1,2,4	
CO-5	Determine the eigen values, the corresponding eigen vectors and diagonalize the given square matrix	1,2,4	1
CO-6	Apply the Rayleigh power method to find the largest eigen value and the corresponding eigen vector.	1,2,4	1

BLOOM'S LEVEL OF THE COURSE OUTCOMES

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO-1	√	√	√			
CO-2	√	√	√	√	√	
CO-3	√	√	√			
CO-4	√	√	√			
CO-5	√	√	√		√	
CO-6	√	√	√		√	

COURSE ARTICULATIONMATRIX

CO#/ Pos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO-1	3	2		2								2		
CO-2	3	2		2								2		
CO-3	3	3		3								2		
CO-4	3	3		3								2		
CO-5	3	2		3								2		
CO-6	3	2		3								2		

Note:1-Low,2-Medium,3-High

COURSE CONTENT

THEORY

Contents
Vector Calculus: Velocity, Acceleration, Tangent and normal vectors, Gradient, Divergence, Curl, Solenoidal and Irrotational vectors, Scalar potential, Vector identities (Basic identities). **Differential Calculus**: nth derivatives of standard function (without proof, *simple problems), Leibnitz theorem (without proof)-simple problems, Taylors series and McLaurin's series expansion for a function of one variable (only problems), Indeterminate forms 'solve using L- Hospital's rule, and Gradient descent method-problems. **LinearAlgebra-1:** Echelon form, Normal form of a matrix, Rank of Matrix, Consistency of the system of homogeneous and non-homogeneous linear equations, Gauss elimination to solve system of equations, and LU decomposition method. **Linear Algebra-2:** Linear transformation, Eigen values and Eigen Vectors up to 3*3 matrices, Diagonalizationfor 2*2 matrices, Rayleigh power method to determiner largest Eigen value and the corresponding Eigen vector.

TEXT BOOKS:

1. Theodore Shifrin, "Multi-Variable Calculus and Linear Algebra with Applications", Wiley, 1st edition, Volume 2, 2018.
2. B.S. Grewal, "Higher Engineering Mathematics", Khanna Publishers, 43nd edition, 2015.
3. Erwin Kreyszig, "Advanced Engineering Mathematics", Wiley Publications, 9th edition,2013.
4. Ron Larson, "Multivariable Calculus, Cengage Learning", 10th Edition, 2013.

REFERENCE BOOKS:

1. B.V. Ramana, "Higher Engineering Mathematics", Tata McGraw Hill Publications, 19th Reprint edition, 2013.
2. R.K.Jain and S.R.K.Iyengar, "Advanced Engineering Mathematics", Narosa Publishing House, 4th edition, 2016.
3. Stanley I. Grossman, "Multivariable Calculus, Linear Algebra, and Differential Equations", 2nd edition, Academic Press 1986.

JOURNALS/MAGAZINE:

https://www.sciencedirect.com/journal/linear-algebra-and-its-applications

SWAYAM/NPTEL/MOOCs:

1. https://nptel.ac.in/courses/111/107/111107108/
2. https://nptel.ac.in/noc/courses/noc20/SEM1/noc20-ma07/
3. https://nptel.ac.in/courses/111/106/111106051/

4. https://nptel.ac.in/courses/111/104/111104092/

5. https://nptel.ac.in/courses/111/104/

11110408/ **SELF-LEARNINGEXERCISES:**

1. Vector spaces, Curvilinear co-ordinates: Cylindrical and spherical co-ordinates.

<table>
<tr><td>Course Title</td><td colspan="4">Physics for Computer Science</td><td colspan="2">Course Type</td><td colspan="2">FC</td></tr>
<tr><td>Course Code</td><td>B24AS0106</td><td>Credit</td><td colspan="2">3</td><td colspan="2">Class</td><td colspan="2">I SEMESTER</td></tr>
<tr><td rowspan="5">Course Structure</td><td>LTP</td><td>Credit</td><td>Contact Hours</td><td>Work Load</td><td colspan="2" rowspan="2">Total Number of Classes Per Semester</td><td colspan="2" rowspan="2">Assessment weightage</td></tr>
<tr><td>Lecture</td><td>3</td><td>3</td><td>3</td></tr>
<tr><td>Tutorial</td><td>-</td><td>-</td><td>-</td><td rowspan="2">Theory</td><td rowspan="2">Practical</td><td rowspan="2">CIE</td><td rowspan="2">SEE</td></tr>
<tr><td>Practical</td><td></td><td></td><td></td></tr>
<tr><td>Total</td><td>3</td><td>3</td><td>3</td><td>42</td><td></td><td>50%</td><td>50%</td></tr>
</table>

COURSEOVERVIEW:

This course introduces the basic concepts of Physics and its applications in Computer Science Engineering courses by emphasizing concepts like Construction and working of Laser sand optical Fibers in point-to-point communication. Quantum Mechanics and its application in Quantum Computation and Haptic Technology. The course has basic laws, expressions, and theories which help to increase the scientific knowledge to analyse upcoming technologies. The course also consists of real-time applications and numerical examples, which makes the course interesting and attractive.

COURSEOBJECTIVES:

The objectives of this course are to:

1. Interpret the knowledge of lasers, and optical fibers and their applications in computer science.
2. Exemplifying the fundamentals of quantum mechanics and its applications in quantum computing.
3. Study physics related to computation aspects.
4. Interpret the properties of dielectric materials for sensor applications
5. Understand the fundamentals and importance of haptic technology.

COURSEOUTCOMES(COs):

On successful completion of this course, the student shall be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Explain the principles, operation, and applications of lasers by analyzing interaction of radiation with matter, Einstein's coefficients, and the conditions required for laser action.	1,2	1
CO2	Apply the concepts of polarization mechanisms in dielectric materials, and *relate* them to applications.	1,2	1
CO3	Use light propagation and waveguide principles to design and analyze optical fiber communication systems with appropriate sources and detectors.	1,2	1
CO4	Analyse the de Broglie hypothesis and wave concepts to calculate electron wavelengths, phase and group velocities.	1,2	1
CO5	Apply quantum mechanical principles and Schrödinger's equation to analyse wave functions, energy eigenvalues, and quantum structures.	1,2	1
CO6	Understand the principles of quantum computation and haptic technology to evaluate and design innovative systems for secure information processing and interactive applications.	1,2	1

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1			√			
CO2			√			
CO3		√				
CO4		√	√			
CO5		√				
CO6		√				

COURSE ARTICULATIONMATRIX

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3										1		
CO2	3	2										1		
CO3	3	2										1		
CO4	3	2										1		

CO5	3	2										1		
CO6	3	2										1		

COURSE ARTICULATION MATRIX:

Note: 1-Low,2-Medium,3-High

COURSE CONTENTS:

Contents
Lasers: Laser and the characteristics, Interaction of radiation with matter (induced absorption, spontaneous and stimulated emission); Expression for energy density of radiation at thermal equilibrium in terms of Einstein's coefficients, Conditions for laser operation (population inversion and Meta stable state); Requisites of laser system, semiconductor laser and its application in laser printing. Numerical. **Dielectric materials:** Introduction, Polarization, susceptibility, polarizability, different polarization mechanisms, Piezoelectric, pyroelectric and ferroelectric dielectrics, properties and applications **Optical Fibers:** Construction and light propagation mechanism in optical fibres (total internal reflection and its importance), Acceptance angle, Numerical Aperture (NA), Expression for numerical aperture in terms of core and cladding refractive indices, Condition for wave propagation in optical fibre, V-number and Modes of propagation, Types of optical fibres, Attenuation and reasons for attenuation, Applications: Explanation of optical fibre communication using block diagram, Optical source (LED) and detector (Photodiode) and their applications, Advantages and limitations of optical communications. Numerical. **Concepts leading to quantum mechanics**: De-Broglie hypothesis, Expression for de-Broglie wavelength of an electron in terms of accelerating potential. Phase velocity and group velocity, Relation between phase velocity and group velocity in terms of C, relation between particle and group velocity. Numericals **Quantum Physics**: Need for quantum mechanics, Wave function, properties of wave function and physical significance. Probability density and Normalization of wave function, Schrodinger independent wave equation, Eigen values and Eigen functions. Applications of Schrödinger wave equation – energy; Eigenvalues of a free particle, Particle in one dimensional infinite potential well with numerical examples. Quantum structures, quantum wire, quantum well and quantum dot structures. **Quantum Computation**: Quantum bits and representation of qubits, distinction between classical bit and quantum bit,quantum logic gates; Qubit as a two-level system. Introduction to quantum Information technology and quantum cryptography. Distinction between classical computer and quantum computer; Open challenges in quantum computation. **Introduction to Haptics:** Types and importance of Haptics**.**

Self-Learning Component:

Holograms using LASERs; Ultrasonics; and Statistical Physics for Computing

TEXTBOOKS:

1. William T. Silfvast, "Laser Fundamentals", Cambridge University press, New York, 2nd Ed, 2004

2. Halliday, R. Resnick, and J. Walker, "Fundamentals of Physics",John Wiley and Sons, New York, 10th edition, 2013.
3. R. K. Gaur and S.L. Gupta, "Engineering Physics", Dhanpat Rai Publications (P) Ltd, New Delhi. 10th Ed, 2014.
4. M.N. Avadhanulu and P.G. Kshirsagar, "A textbook of Engineering Physics", S. Chand and Company, New Delhi, 17th Ed, 2014.
5. Lorrain and O. Corson, "EM Waves and Fields",3rd Ed, CBS Publishers, 2003.
6. Dr. Nidhi, Ismail Keshta, Dr. Abhishek Agarwal, Dr. Vandana Rathore, "Quantum computing for beginners", Xoffencer International Publisher, 1st Ed, 2023.
7. Martin Grunwald, "Human Haptic Perception Basics and Applications", Springer Basel AG, 1st Ed, 2008.
8. R.A., Serway, John W. Jewett, Physics for Scientists and Engineers, Technology, Lachina publishing Services, 9th Ed, 2015.

REFERENCEBOOKS:

1. Arthur Beiser, "Concepts of modern Physics", Tata McGraw Hill publications, NewDelhi, 8th Ed, 2011.
2. S. O. Pillai, "Solid State Physics", New Age International publishers, New Delhi, 9th Ed, 2010.
3. Michael A. Nielsen, Isaac L. Chuang," Quantum Computation and Quantum Information",Cambridge University Press, 1st Ed, 2010.

Additional material:

1. LASER: https://www.youtube.com/watch?v=WgzynezPiyc
2. Optical Fiber : https://www.youtube.com/watch?v=N_kA8EpCUQo
3. Polarization: https://www.youtube.com/watch?v=KBJl1qiYOgo
4. Quantum Mechanics : https://www.youtube.com/watch?v=p7bzE1E5PMY&t=136s
5. Quantum Computing : https://www.youtube.com/watch?v=jHoEjvuPoB8
6. Quantum Computing :https://www.youtube.com/watch?v=ZuvCUU2jD30
7. Statistical Physics Simulation : https://phet.colorado.edu/sims/html/plinko-probability/latest/plinkoprobability_en.html
8. NPTEL Quantum Computing : https://archive.nptel.ac.in/courses/115/101/1151010

Course Title	**Fundamentals of Mechanical Engineering**				**Course Type**		**HC**	
Course Code	**B24ME0105**	**Credit**	**3**		**Class**		**I Semester**	
Course Structure	LTP	Credit	Contact Hours	Work Load	Total Number of Classes Per Semester		Assessment Weightage	
	Lecture	3	3	3				
	Tutorial	-	-	-	Theory	Practical	**CIE**	**SEE**
	Practical	-	-	-				
	Total	**3**	**3**	**3**	**42**	**-**	**50%**	**50%**

COURSE OVERVIEW:

This course is designed to understand the concepts of green energy, hybrid vehicles, IC engines, refrigeration and air condition system, mechatronics, control systems, advanced manufacturing and automation. Further, this course includes applications of python programming for mechanical engineering. The students will be imparted to evaluate the performance of IC engines, use of machine elements like, belt drives, chain drives and gear drives. Students are introduced to the modern manufacturing technologies like CNC, 3D printing and applications of IIoT, Machine learning and robotics.

COURSE OBJECTIVES:

The objectives of this course are:

1. To develop the basic knowledge on energy resources, green energy concept, IC engines, turbines and refrigeration and air conditioning.
2. To incorporate the concept of different types of machine elements such as, gear drives, belt drives & chain drives.
3. To incorporate the concepts of mechatronics and robotics systems, modern manufacturing processes like CNC, additive manufacturing and digital manufacturing technology and its applications.
4. To provide a basic understanding role of computer science in mechanical engineering applications.
5. To provide the basic understanding of coding for mechanical engineering applications.

COURSE OUTCOMES (COs):

On successful completion of this course, the student shall be able to:

CO	Course Outcomes	POs	PSOs
CO1	Apply the knowledge of energy resources to analyze real-world energy challenges and propose potential solutions and understanding of the concept of sustainability in the context of energy resources.	1, 2	1,2
CO2	Develop a comprehensive understanding of the working principles and application underlying prime movers and evaluate the performance of IC engines using python programming.	1, 2, 5	1,2
CO3	Gain a solid understanding of the fundamental principles refrigeration and air conditioning systems and develop a comprehensive understanding of working mechanical drives such as gears, belts, chains.	1	1,2
CO4	Identify the key elements of mechatronics system and interface sensors / transducer output with microprocessor for design or controlling of the system.	1, 5	1
CO5	Gain a comprehensive understanding of the principles and concepts underlying Industry 4.0, including, the Internet of Things (IoT), Machine Learning (AI), 3D printing and digital technologies.	1, 5	1,2,3
CO6	Understand the applications of robotics in various fields, such as manufacturing, agriculture, logistics, aerospace, etc and understand the concept of measurement system and control system used in robotics.	1	1,2,3

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

CO	Bloom's Level					
	Remember (L1)	**Understand (L2)**	**Apply (L3)**	**Analyze (L4)**	**Evaluate (L5)**	**Create (L6)**
CO1			√			
CO2			√			
CO3		√				
CO4		√	√			
CO5		√				
CO6		√				

COURSE ARTICULATION MATRIX:

CO / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1										2	1	
CO2	2	2			1							2	1	
CO3	2											2	1	
CO4	2				1							2		
CO5	2				1							2	1	1
CO6	2											2	1	1
Average	**2**	**1.5**			**1**							**2**	**1**	**1**

Note: 1-Low, 2-Medium, 3-High

CONTENTS

COURSE CONTENT:

Energy Resources: Types of energy resources: Green energy concept – Hydrogen energy; production using electrolysis process, advantages and disadvantages. Solar P-V system and wind energy.

IC Engines: Fuels and properties, classification of IC engines, parts and terminology, working of 4-stroke petrol engine, numerical on IP, BP, FP and Mechanical Efficiency**,** *python programming for evaluation of engine performance.*

Concept of electric and hybrid vehicles.

Gas Turbines: Open and closed cycle gas turbine, hydraulic turbines, Classification and working of Pelton wheel.

Refrigeration and Air Conditioning: Principle of Refrigeration system, working of domestic refrigerator (VCR) and window air conditioner.

Machine Elements: Belt drives, open and crossed belt drive, gear drives and chain drives, numerical on velocity ratio, p*ython programming for computation of speed ratio.*

Mechatronics Systems: Definition of mechatronics, components of mechatronics, basic Terminologies, Open loop and Closed loop control systems, microprocessor based control systems, static characteristics of sensor, working of capacitance sensor, eddy current sensor, hall effect sensor-Light sensors, optical encoders.

Introduction to Advanced Manufacturing and Industry 4.0 -Computer numerical control (CNC) machines, laser engraving, 3D printing, classification, applications, working of FDM, Introduction to Digital manufacturing- machine learning (ML) applications, Industrial Internet of Things (IIoT), Human Machine Interface.

Robotics and Automation: Need of Automated Guided Vehicle (Retrieval/storage system) in industries, applications of Robots, measurement system, open and closed loop control system. Adoption of logic gates in automation.

Self-Learning Component: Hybrid vehicles, Engineering Materials, Need of Automated Guided Vehicle. Basics of numerical python and MAT Plot.

TEXT BOOKS:

1. K R Gopala Krishna, Sudheer Gopala Krishna and S C Sharma, "Elements of Mechanical Engineering", Subhash Publishers, 13^{th} Edition, 2015.

2. Roy & Choudhury, "Elements of Mechanical Engineering", Media Promoters & Publishers Pvt. Ltd, 2000.

3. Manaranjan Pradhan and Dinesh Kumar U, "Machine Learning using Python", Wiley Publishers, 1^{st} Edition, 2019

REFERENCE BOOKS:

1. SKH Chowdhary, AKH Chowdhary and Nirjhar Roy, "The Elements of Workshop Technology - Vol I & II", Media Promotors and publisher, 11^{th} Edition, 2001.

2. William Bolton, "Mechatronics Electronics Control Systems in Mechanical and Electrical Engineering", Pearson, 2015.

3. K. K. Appukuttan, “Introduction to Mechatronics”, Oxford University Press, 2007.
4. Madhumathy P, M Vinoth Kumar, R. Umamaheswari “Machine Learning and IoT for Intelligent Systems and Smart Applications” CRC Press publisher, 2021.
5. John V. Guttag, “Introduction to Computation and Programming using Python”, MIT Press, 2016.

JOURNALS/MAGAZINES

1. International Journal of Refrigeration.
2. https://www.journals.elsevier.com/international-journal-of-hydrogen-energy
3. https://www.researchgate.net/publication/369942210_Fuel_Cell_Electric_Vehicle_Modeling_Using_HybridBatteryFuel_Cell_Vehicle_Modeling_Tool_HBFCMT.

SWAYAM/NPTEL/MOOCs:

1. https://www.coursera.org/browse/physical-science-and-engineering/mechanical-engineering
2. https://www.my-mooc.com/en/categorie/mechanical-engineering

<table>
<tr><td>Course Title</td><td colspan="4">Business Analysis & Software Design</td><td colspan="2">Course Type</td><td colspan="2">SE</td></tr>
<tr><td>Course Code</td><td>B25CS0201</td><td>Credits</td><td colspan="2">2</td><td colspan="2">Class</td><td colspan="2">I Semester</td></tr>
<tr><td rowspan="5">Course Structure</td><td>LTP</td><td>Credits</td><td>Contact Hours</td><td>Workload</td><td colspan="2" rowspan="3">Total Number of Classes Per Semester</td><td colspan="2" rowspan="3">Assessment Weightage</td></tr>
<tr><td>Lecture</td><td>2</td><td>2</td><td>2</td></tr>
<tr><td>Tutorial</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>Practice</td><td>-</td><td>-</td><td>-</td><td>Theory</td><td>Practical</td><td>CIE</td><td>SEE</td></tr>
<tr><td>Total</td><td>2</td><td>2</td><td>2</td><td>26</td><td>0</td><td>50%</td><td>50%</td></tr>
</table>

COURSE OVERVIEW:

This foundation course introduces first-semester students to systems thinking, writing clear specifications, and basic design diagrams using simple open-source tools. Students learn to describe, analyze, and improve real-world systems like UPI, Ola, Swiggy, and College Bus apps. The emphasis is on thinking, clarity, and structured problem-solving rather than coding.

COURSE OBJECTIVE (S):

By the end of this course, students will:

1. Understand the concept of systems thinking and identify key components such as actors, stakeholders, goals, and constraints in everyday digital applications.
2. Develop skills to write structured and verifiable specifications, including user stories, functional and non-functional requirements, and acceptance criteria.
3. Apply modular thinking and structured decomposition to represent system architecture using basic UML models.
4. Construct system diagrams such as use case, sequence, and flow diagrams using open-source visual modeling tools like draw.io and Mermaid.
5. Utilize prompt engineering and structured problem-solving techniques to assist in drafting requirement and design documents using AI-powered tools.
6. Analyze and improve real-world software systems by creating simplified design representations and communicating

solutions through visual diagrams and specifications.

COURSE OUTCOMES (COs):

After the completion of the course, the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Explain what a system is and identify stakeholders, use cases, and requirements.	1, 2, 10	1-3
CO2	Write simple but detailed specifications (scope, user stories, functional & non-functional requirements, acceptance criteria).	2, 3, 10, 11	1-3
CO3	Apply modular thinking and separation of concerns to represent system components.	1, 2, 3, 5	1-3
CO4	Construct simple diagrams (use case, block, sequence/flow) using draw.io (Mermaid) or StarUML or PlantUML.	3, 4, 5	1-3
CO5	Use prompt engineering to assist in drafting specifications and design ocuments.	2, 3, 5	1-3
CO6	Analyze a familiar system and propose improvements through a specification + design demo.	2, 3, 4, 5, 9, 10, 11	1-3

BLOOM'S LEVEL OF THE COURSE OUTCOMES

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1		√				
CO2			√			
CO3			√			
CO4			√			
CO5			√			
CO6				√		√

COURSE ARTICULATION MATRIX

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2								2				
CO2		2	3								3			
CO3	2	2	3		2									
CO4			2	2	3									
CO5		2	2		3									
CO6		3	3	2	2				2	2	2			

Note:1-Low,2-Medium,3-High **COURSE CONTENT**

THEORY:

Contents
Systems Thinking & Requirements: What is a system? Examples: UPI, Ola. Stakeholders, goals, and use cases. Writing structured requirements with Specification Template (Scope, Actors, User Stories, FRs, NFRs, Acceptance Criteria). Activity: Write 5–6 requirements for UPI; create a simple sequence diagram in draw.io (Mermaid) **Modular Thinking & UML Basics:** Separation of Concerns (SoC) / modular decomposition. UML basics: Use Case & Block/Component diagrams. Example: Decomposing UPI/Ola into modules (Auth, Payment, History, Notifications).Activity: Use draw.io or StarUML/ PlantUML to create a use-case + block diagram **Prompt Engineering for Specifications & Design:** Introduction to AI & LLMs (conceptual overview).Basics of Prompt Engineering (zero-shot, few-shot, chained prompts)..How prompts can scaffold: User Stories → Requirements → Acceptance Criteria → Draft Design. Activity: Use prompts to generate draft specifications for UPI and refine manually **Familiar Applications & Project Demo:** Case study: College Bus / Swiggy. Identify problems; propose improvements. Group Project: Write specifications (≤3 pages) Draw Use Case + Sequence/Flow diagrams in draw.io (Mermaid) Mock UI screens using Bolt.new or Rocket.new (optional) Final Project Demo: Each student/group demonstrates their project (spec + diagrams + UI)

TEXTBOOKS:

1. **“Software Engineering: A Practitioner’s Approach”** – *Roger S. Pressman*
2. **“Object-Oriented Modeling and Design with UML”** – *Michael Blaha & James Rumbaugh*

REFERENCE BOOKS:

1. **“Business Analysis”** – *Debra Paul, James Cadle (BCS Business Analysis Series)*
2. **“Head First Design Patterns”** – *Eric Freeman, Elisabeth Robson*→ Easy to read, visual book on modular thinking, separation of concerns, and reusable solutions.

JOURNALS/MAGAZINES:

1. **IEEE Software Magazine**
2. **ACM Communications (Education & Software Sections)**

SWAYAM/NPTEL/MOOCs:

1. SWAYAM – “Software Engineering”
   Instructor: Prof. Rajib Mall (IIT Kharagpur)
   https://onlinecourses.nptel.ac.in/noc23-cs58
2. NPTEL – “System Analysis and Design”
   Instructor: Prof. Anirban Mondal (IIT Roorkee)
   https://nptel.ac.in/courses/106105182

Course Title	**Introduction to C Programming**				**Course Type**		**HC**	
Course Code	B25CI0101	**Credits**	**2**		**Class**		**I Semester**	
Course Structure	LTP	Credits	Contact Hours	Work Load	Total Number of Classes Per Semester		Assessment Weightage	
	Lecture	1	1	1				
	Tutorial	1	2	2				
	Practice	-	-	-	Theory	Practical	CIE	SEE
	Total	**2**	**3**	**3**	**42**	**-**	**50%**	**50%**

COURSE OVERVIEW:

This course is designed to introduce first-year B.Tech students to the foundational concepts of computer programming using the C language. It begins with an orientation to basic operating system usage (Unix and Windows) and develops the ability to express solutions through flowcharts and algorithms. The course then guides students through essential C programming constructs such as variables, data types, input/output functions, and control flow statements. Emphasis is placed on understanding how to represent and manipulate data using arrays, strings, and structures, including array of structures. Students will also be introduced to modular programming through user-defined and recursive functions. Through activity-based learning approach and guided practice, the course aims to build a strong foundation in computational thinking and structured programming essential for problem-solving in engineering contexts.

COURSE OBJECTIVE (S):

The objectives of this course are to:

1. To introduce students to the fundamentals of problem-solving and the use of flowcharts and algorithms for representing solutions.
2. To familiarize students with basic Unix and Windows operating system commands essential for creating, compiling, debugging, and executing C programs.
3. To develop an understanding of foundational C programming concepts such as variables, data types, and input/output

operations.
4. To enable students to apply conditional and looping control flow constructs for solving illustrative problems.
5. To build proficiency in handling data through arrays, strings, structures, and arrays of structures in C.
6. To cultivate the ability to design modular programs using user-defined and recursive functions for efficient problem-solving.

COURSE OUTCOMES (COs)

After the completion of the course, the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Write flowcharts and algorithms to solve simple problems.	1-3	1
CO2	Understand basic usage of Unix and Windows operating system required for creating, compiling, debugging and running C programs.	1-5	1
CO3	Understand foundational concept of C programming constructs such as variables, data types, and standard I/O functions for solving simple problems.	1-5	1
CO4	Understand conditional and unconditional control flow statements including if, for and while using illustrative example problems.	1-3,5	2,3
CO5	Understand arrays, strings, structures and array of structures using illustrative C programs.	1-5	2,3
CO6	Understand functions to achieve modularity using illustrative C programs.	1-5	2,3

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

CO#	Bloom's Level					
	Remember(L1)	Understand(L2)	pply (L3)	nalyze (L4)	aluate (L5)	reate (L6)
CO1		✓				
CO2		✓				
CO3			✓			
CO4			✓			
CO5		✓	✓			
CO6		✓				

COURSE ARTICULATIONMATRIX:

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1	3									3		
CO2	1	3	2	2	2							3		
CO3	2	2	2		1								3	3
CO4	3	3	3	1	1								3	3
CO5	3	3	3	2	2									
CO6	3	3	3	2	2				3			3	3	2

Note: 1-Low, 2-Medium, 3-High

COURSE CONTENT THEORY

Content
Introduction to C Programming Constructs: Introduction to algorithm and flowchart, overview of C programming language and structure of a C program. Variables, expressions and data types, Input and output functions. **Operators in C:** Arithmetic operators, Relational operators, Logical operators, Bitwise operators, Assignment operators, Conditional operator, Precedence and associativity of operators and Type conversion. **Control Flow in C:** Conditional control statements: if, if-else, else-if ladder and switch case. Looping statements: for, nested loops, while and do-while. Unconditional Statements: continue, return, and break. **Data Structures in C – Arrays, Strings, and Structures:** One and two-dimensional arrays. String handling in C: character arrays, usage of standard library functions. Structure and Union: definition, declaration, and accessing members. Array of structures. **Modular Programming with Functions:** Concept of modularity and need for functions, User-defined functions: function declaration, definition, and call. Function arguments and return values (call by value). **Unix and windows Commands:** Installation of required software. Commands for setting path variables, creating directory, changing working directory, copying files, deleting files.

TEXT BOOKS:

1. Herbert Schildt, "C: The Complete Reference", 4^{th} edition, TATA McGraw Hill, 2000.
2. B.W. Kernighan & D.M. Ritchie, "C Programming Language", 2^{nd} Edition, PRENTICE HALL SOFTWARE SERIES, 2005.
3. B.S. Anami, S.A. Angadi and S. S. Manvi, "Computer Concepts and C Programming: A Holistic Approach", second edition, PHI,2008.
4. Yashavant Kanetkar, "Let us C" , BPB , 19^{th} edition, 2022

REFERENCE BOOKS:

1. Balaguruswamy," Programming in ANSI C", 4^{th} edition, TATA MCGRAW Hill, 2008.
2. Donald Hearn, Pauline Baker," Computer Graphics C Version", second edition, Pearson Education, 2004.
3. **Jon Bentley, "**Programming Pearls", 2^{nd} edition, PEARSON,2002.

JOURNALS/MAGAZINES:

1. https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=6294 (IEEE Journal/Magazine on IT Professional)
2. https://ieeexplore.ieee.org/document/1267572 (IEEE Computing in Science and Engineering)

SWAYAMNPTEL/MOOCs

1. https://onlinecourses.nptel.ac.in/noc20_cs06/preview (Problem Solving through Programming in C)
2. https://www.edx.org/course/c-programming-getting-started (C Programming Getting started)
3. https://www.coursera.org/specializations/c-programming (Introduction to C programming)

Self-Learning Component:

1. GitHub and Raptor
2. Visualize C programming using pythontutor.com
3. USB Live Linux
4. GDB

Course Title	**Internet of Things**				**Course Type**		**HC(Integrated)**	
Course Code	**B24EN0101**	**Credit**	**2**		**Class**		**I/II Semester**	
Course Structure	LTP	Credit	contact Hours	Work Load	Total Number of Classes Per Semester		Assessment Weightage	
	Lecture	1	1	1				
	Tutorial	-	-	-	Theory	Practical	**CIE**	**SEE**
	Practical	1	2	2				
	Total	**2**	**3**	**3**	**14**	**28**	**50%**	**50%**

COURSE OVERVIEW:

The Internet of Things (*IoT)* expands access to the world-wide web from computers, smartphones, and other typical devices to create a vast network of appliances, toys, apparel, and other goods that are capable of connecting to the Internet. This introductory course focuses on IoT architecture, Boards, sensors, actuators and communication protocols. The course is supported with hands on sessions that incorporates different types of sensors interfaced with IoT board to build IoT projects to solve real time problems. The case study of deployment of IoT in various applications are provided.

COURSE OBJECTIVES:

The objectives of this course are :

1. To understand the fundamentals of Internet of Things.
2. Inculcate knowledge of IoT devices, Sensors and Communication Protocols in various application domains.
3. Gain expertise in interfacing of various sensors to IoT Boards, Arduino and Node MCU.
4. Discuss various applications of IoT.

COURSE OUTCOMES (COs):

On successful completion of this course, the student shall be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Describe IoT architecture and Models of IOT	1,2,3,4,5	1,2
CO2	Identify and work on different IoT development boards	1,2,3,4,5	1,2
CO3	Identify communication technologies, protocols, and cloud services	1,2,3,4,5	1,2
CO4	Identify sensors & actuators and demonstrate the interfacing of sensors & actuators to IoT boards	1,2,3,4,5,9,10	1,2
CO5	Interpret various Applications of IoT	1,2,3,4,5,6,9,10	1,2,3
CO6	Develop simple IoT projects and modules	1,2,3,4,5,9,10,11	1,2,3

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1	√	√				
CO2	√	√				
CO3	√	√	√			
CO4	√	√	√	√		
CO5	√	√	√	√		
CO6	√	√	√	√		

COURSE ARTICULATION MATRIX:

CO#/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	1	1							3	3	
CO2	2	3	1	1	1							3	3	
CO3	3	2	1	1	2							2	2	
CO4	3	3	3	2	2				2	1		2	2	
CO5	3	2	2	1	2	2			2	1		2	2	1
CO6	3	2	2	1	2				2	2	2	2	2	2

Note:1-Low,2-Medium,3-High

COURSE CONTENT

Contents
IoT Basics and Arduino UNO Board Introduction to IoT and its working, IoT Architecture, Models of IOT, Characteristics of IoT, Merits, demerits, futures scope and challenges of IoT, IOT Echo system, Arduino UNO board, memories, microcontroller, Basic programming of Arduino UNO board: Numeric data types, text data types, Array data types, Digital input output functions, Communication Technologies: Bluetooth, ZigBee, Wi-Fi, Cellular **IoT Enabling Technologies and Applications** IoT Development Boards: Node MCU, Raspberry Pi; Sensors and Actuators: Temperature Sensor, PIR Sensor, Ultrasonic senor, LDR, Applications of IoT, Protocols: HTTP, MQTT, CoAP; IoT Cloud Platforms: Arduino Cloud, Thing Speak, Blink Cloud

Experiments:

Sl. No.	Name of the Practice Session	Tools and Techniques	Expected Skill /Ability
	Arduino UNO		
1	**Experiment-01** Introduction to Arduino Board & getting started with Arduino IDE software. Write a program to blink an LED: a) Infinite number of times with ON & OFF duration of 1 sec b) Only 3 times with ON and OFF duration 2 sec	Hardware & software for Arduino UNO Arduino UNO Board Arduino IDE LED	Identifications of various parts of Arduino UNO and Coding
2	**Experiment-02** a) Write a program to increase and decrease the brightness of LED. b) Write a program to control the brightness of LED using Potentiometer.	Arduino UNO Arduino IDE LED Potentiometer	Interface of Potentiometer
3	**Experiment-03** a) Write a program to interface LDR to Arduino board and display the voltage across LDR on serial monitor. b) Write a program to control the brightness of LED based on the intensity of light on LDR.	Arduino UNO Arduino IDE LED LDR	Interface LDR sensor
4	**Experiment-04** a) Write a program to interface temperature sensor and display the values on the serial monitor. b) Write a program to display the range of temperature on LCD.	Arduino UNO Arduino IDE LCD Temperature sensor	Interface Temperature sensor
5	**Experiment-05** Write a program to interface ultrasonic sensor and display the distance from an object.	Arduino UNO Arduino IDE Ultrasonic sensor	Interface Ultrasonic sensor
	NODU MCU		
6	**Experiment-06** Introduction to NODE MCU and Write a Program for 2-BIT COUNTER using Node MCU.	Hardware & software for NODE MCU NODE MCU Board Two LED’s	Identifications of various parts of NODE MCU and Coding
7	**Experiment-07** Write a program to interface motion sensor and display its status using LED. If motion is detected,	NODE MCU LED PIR sensor	Interface PIR sensor

	turn on LED otherwise keeps the LED off.		
8	**Experiment-08** Write a program to interface Single LED blinking using Node MCU and Bluetooth module.	Node MCU Bluetooth module LED	Connection of Bluetooth to NODE MCU
9	**Experiment-09** Write a program to control the status of LED using relay.	Node MCU Bluetooth module LED	Control of LED through Relay
10	**Experiment-10** Write a program to interface temperature sensor and display the values on Web Page/Cloud.	Node MCU Temperature Sensor LED Cloud Service	Connect to cloud and display the values
Part-B (Case Study/ Projects - Sample Topics)			

IoT based Automated Table Lamp **IoT based Light Dimmer and Speed Controller** **IoT based Energy Monitor and Over Current Cut-off** **IoT based Smart Home Controller Using Blynk** **IoT based Motion Detector Using Cayenne** **IoT based Air Pollution Meter** **IoT based Smart Camera** **IoT based Pet Feeder** **IoT based Electronic Door Opener** **IoT based Underground Cable Fault Detector** **IoT based Air & Sound Pollution Monitoring System** **IoT based Weather Reporting System** **IoT based Toll Booth Manager System** **IoT based Heart Attack Detection & Heart Rate Monitor** **IoT based Person/Wheelchair Fall Detection** **IoT based Water Quality Monitoring System**	**IoT based Patient Health Monitoring** **IoT based Garbage Monitoring System** **IoT based Liquid Level Monitoring System** **IoT based Biometric Attendance System** **IoT based Irrigation Monitoring & Controller System** **IoT based Gas Pipe Leakage Detector** **IoT based Alcohol & Health Monitoring System** **IoT based Streetlight Controller System** **IoT based Traffic Signal Monitoring & Controller System** **IoT based Fire Department Alerting System** **IoT based Antenna Positioning System** **IoT based Garbage Monitoring with Weight Sensing** **IoT based Colour Based Product Sorting Machine** **IoT based Smart Mirror with News & Temperature** **IoT based Car Parking System** **IoT based Automatic Vehicle Accident Detection and Rescue System**

	Challenge Experiments
1	a) Introduction to ESP module & programming using Arduino IDE software b) Write a program to demonstrates how to use Wifi module ESP8266-01 to blink LED (with simple LED)
2	Write a program to demonstrate how ESP8266 can be used as an HTTP client and HTTP server to control and monitor the status of an LED
3	Write a program demonstrate how ESP8266 can be used as HTTP Webserver and get commands from the client (mobile/Laptop) directly.
4	Write a program to demonstrate how to implement Publisher/Subscriber method (MQTT) to

A	control and monitor the ESP8266 GPIO2 LED	
B	Write a program to demonstrate how ESP8266 can be used to log sensor data into thing speak cloud.	
C	Viva questions	
D	Datasheets	

TEXT BOOKS:

1. Vijay Madisetti, Arshdeep Bahga , "Internet of Things: A Hands-On- Approach " Second edition 2014, ISBN: 978 0996025515.

REFERENCE BOOKS:

1. Raj Kamal," Internet of Things: Architecture & design Principle", McGraw Hill Education 2017.

SWAYAM/NPTEL/MOOCs/Additional Materials:

1. https://www.coursera.org/learn/iot
2. https://www.coursera.org/learn/interface-with-arduino
3. https://www.coursera.org/learn/iot
4. https://www.coursera.org/learn/interface-with-arduino

Course Title	**Introduction to C Programming Lab**				**Course Type**		**HC**	
Course Code	B25CI0102	**Credits**	1		**Class**		**I Semester**	
COURSE STRUCTURE	LTP	Credits	Contact Hours	Work Load	Total Number of Classes Per Semester		Assessment in Weightage	
	Lecture	-	-	-				
	Tutorial	-	-	-	Theory	Practical	IA	SEE
	Practice	1	2	2				

	Total	1	2	2	-	28	**50%**	**50%**

COURSE OVERVIEW:

This course is designed to introduce first-year B.Tech students to the foundational concepts of computer programming using the C language. It begins with an orientation to basic operating system usage (Unix and Windows) and develops the ability to express solutions through flowcharts and algorithms. The course then guides students through essential C programming constructs such as variables, data types, input/output functions, and control flow statements. Emphasis is placed on understanding how to represent and manipulate data using arrays, strings, and structures, including array of structures. Students will also be introduced to modular programming through user-defined and recursive functions. Through activity-based learning approach and guided practice, the course aims to build a strong foundation in computational thinking and structured programming essential for problem-solving in engineering contexts.

COURSE OBJECTIVE (S):

1. To introduce students to the fundamentals of problem-solving and the use of flowcharts and algorithms for representing solutions.
2. To familiarize students with basic Unix and Windows operating system commands essential for creating, compiling, debugging, and executing C programs.
3. To develop an understanding of foundational C programming concepts such as variables, data types, and input/output operations.
4. To enable students to apply conditional and looping control flow constructs for solving illustrative problems.
5. To build proficiency in handling data through arrays, strings, structures, and arrays of structures in C.
6. To cultivate the ability to design modular programs using user-defined and recursive functions for efficient problem-solving.

COURSE OUTCOMES (COs)

On successful completion of this course; the student shall be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Write flowcharts and algorithms to solve simple problems.	1-3	1
CO2	Understand basic usage of Unix and Windows operating system required for creating, compiling, debugging and running C programs.	1-5	1
CO3	Understand foundational concept of C programming constructs such as variables, data types, and standard I/O functions for solving simple problems.	1-5	1
CO4	Understand conditional and unconditional control flow statements including if, for and while using illustrative example problems.	1-3,5	2,3
CO5	Understand arrays, strings, structures and array of structures using illustrative C programs.	1-5	2,3
CO6	Understand functions to achieve modularity using illustrative C programs.	1-5	2,3

BLOOM'S LEVELOF THECOURSE OUTCOMES

CO#	Bloom's Level					
	Remember(L1)	Understand(L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1		✓				
CO2		✓				
CO3			✓			
CO4			✓			
CO5		✓	✓			
CO6		✓				

COURSE ARTICULATIONMATRIX

CO#/ Pos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1	3						3	3		3		
CO2	1	3	2	2	2				3	3		3		
CO3	2	2	2		1				3	3			3	3
CO4	3	3	3	1	1				3	3			3	3
CO5	3	3	3	2	2				3	3				
CO6	3	3	3	2	2				3	3		3	3	2

Note:1-Low,2-Medium,3-High

No.	Title of the Experiment
1	Write algorithms and flowchart for basic programming tasks using Raptor.
2	Explore basic Unix commands for creating, compiling, debugging and running C programs.
3	Write programs to perform basic operations using variables and expressions.
4	Write programs using if-else statements to perform comparison operations.
5	Write programs using for loop to compute aggregate values.
6	Write programs involving numerical sequences using while loop.
7	Write programs using arrays to perform aggregate and search operations.
8	Write programs for basic string processing.
9	Write programs using two-dimensional arrays to perform matrix operations.
10	Write programs involving variables and expressions using structures.
11	Write programs using array of structures to perform CRUD operations.
12	Write programs using functions to perform basic operations using variables and expressions.

TEXT BOOKS:

1. B.W. Kernighan & D.M. Ritchie, "C Programming Language", 2nd Edition, PRENTICE HALL
2. SOFTWARE SERIES, 2005.
3. Herbert Schildt, “C: The Complete Reference”, 4th edition, TATA McGraw Hill, 2000.
4. B.S. Anami, S.A. Angadi and S. S. Manvi, “Computer Concepts and C Programming: A
5. Holistic Approach”, second edition, PHI,2008.

REFERENCE BOOKS:

1. Balaguruswamy,” Programming in ANSI C”, 4th edition, TATA MCGRAW Hill, 2008.
2. Donald Hearn, Pauline Baker,“ Computer Graphics C Version”, second edition, Pearson Education, 2004.

Course Title	**Physics for Computer Science Lab**				**Course Type**		**HC**	
Course Code	**B24AS0108**	**Credits**	**1**		**Class**		**I Semester**	
Course Structure	**LTP**	**Credits**	**Contact Hours**	**Workload**	**Total Number of Classes Per Semester**		**Assessment in Weightage**	
	Lecture	-	-	-				
	Tutorial	-	-	-	**Theory**	**Practical**	**CIE**	**SEE**
	Practice	1	2	2				
	Total	**1**	**2**	**2**	**-**	**28**	**50%**	**50%**

COURSE DESCRIPTION:

Engineering Physics is very important and necessary basic subject for all branches of engineering students. It provides the fundamental knowledge of basic principles of Physics which is required for basic foundation in engineering education irrespective of branch. This course introduces the experimental concepts of Physics and its applications to Computer Science Engineering courses by emphasizing the following concepts: electrical properties, dielectrics, semiconductors, LASERs, and optical properties. This course provides a basic understanding of the working of different electronic components. The course also emphasizes simulating the working of some electronic components.

COURSE OBJECTIVE(S):

The objectives of this course are to:

1. Demonstrate the principles covered in your study material in physics.
2. Provide familiarity with apparatus and enable them to handle the instruments and apparatus with purpose.
3. Identify the process to study conditions for a given experiment.
4. Develop an attitude of perfection in practical tasks.
5. Simulate the working of different electronic components using mobile or a computer.

COURSE OUTCOMES (COs)

After the completion of the course the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Constructing simple circuits and performing experiments to study voltage-current response.	1,2,3,4,8	1,2,3
CO2	Determination of unknown physical parameters related to LASERs	1,2,3,5,8	2, 3
CO3	Determination of unknown physical parameters related to optical fibers.	2,3, 4,7,8	1, 2, 3
C04	Determine the dielectric constant of the material.	2,3, 4,7,8	1, 2, 3
CO5	Simulating the experimental data of physical experiments	1 to 5,8	1, 2, 3
CO6	Simulate the working of electronic circuits	1 to 5, 7, 8	1,2, 3

BLOOM'S LEVEL OF THE COURSE OUTCOMES

CO#	Bloom's Level					
	Remember (L1)	**Understand (L2)**	**Apply(L3)**	**Analyze(L4)**	**Evaluate(L5)**	**Create(L6)**
CO1				✓		
CO2			✓	✓		
CO3			✓	✓		
C04			✓	✓	✓	✓
CO5				✓	✓	
CO6			✓	✓	✓	✓

COURSE ARTICULATION MATRIX

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	1	1				3						1
CO2	3	3	1		1			3						2
CO3		2	1	2			2	3						1
CO4		2	1	2			2	3						1
CO5	3	2	1	2	1			3						2
CO6	3	2	1	2	1		2	3						1

List of Experiments

No.	Title of the Experiment	Tools and Techniques	Expected Skill/Ability
1	Light-dependent characteristics of a photosensor.	Photodiode, Ammeters, and voltmeters.	Circuit construction, Perform, and plotting of data using Matlab/origin
2	Determination of wavelength of IR LEDs and energy gap in applications to remote controllers.	LEDs, AC source, and voltmeters.	Circuit construction, Perform, and plotting of data using Mat lab/origin
3	Simulation of experimental data using Origin software	Origin software	Simulated graphs
4	Determination of numerical aperture and acceptance angle to launch the optical power into the optical fiber for digital communication.	LASER. Optical fiber light source, DC supply	Experimental setup, plotting of data using Matlab/origin
5.	Determination of the attenuation coefficient of the given optical fiber for long-distance communication.	LASER. Optical fibers light source, DC supply	Experimental setup, plotting of data using Matlab/origin
6	Determination of wavelength of LASER for LASER printing and digital communications.	LASER. Optical fiber light source, DC supply	Experimental setup, plotting of data using Matlab/origin
7	Verification of splicing loss experimental for optical fiber.	Diode laser, digital dc micrometer two OFC (1.5m &2.5m), optical sensor	Circuit construction, Perform, and plotting of data
8	Determination of dielectric constant using charging and discharging of capacitor	Capacitor, resistor, voltmeters	Circuit construction, Perform, and plotting of data
9	Study of motion using spreadsheets	Simulation software	Circuit construction, Perform, and plotting of data
10	PhET interactive simulations (Projectile motion, wave-particle dualism).	Simulation software	Circuit construction, Perform, and plotting of data

Part B: Demo and Simulation

No.	Title of the Experiment	Tools and Techniques	Expected Skill/Ability
1	Generation of ultrasonic waves by piezoelectric transducers and its velocity determination in liquid for the Haptics applications.	piezoelectric transducers	To understand the concept of Haptics
2	Demonstration of superposition of waves of monochromatic light.	Oscillators and CRO	Interference of waves to understand the quantum superposition
Experiments for the mathematical model			
3	Bending of a single cantilever, its mathematical model. Time-period calculation of simple pendulum, its mathematical model.	Every circuit (android app) Tina (Online simulator)	Visualize, simulate, and analyze

Text Books:

1. M.N. Avadhanulu and P.G. Kshirsagar, "A Textbook of Engineering Physics", S. Chand & Company Ltd, New Delhi, 10thEd, 2008.
2. Gaur and Gupta, "Engineering Physics",Dhanpat Rai Publications, 6thEd, 2017.
3. S.P. Singh, "Advanced Practical Physics Vol.- I", Pragati Prakashan, 15th Ed, 2017.

4. G.L.Souires, “Practical Physics”, Cambridge University, UK, 4th Ed,2001.

Reference Books:

1. Arthur Beiser, “Concepts of Modern Physics”, Tata McGraw Hill Edu Pvt. Ltd- New Delhi, 6thEd, 2006.
2. S O Pillai, “Solid State Physics”, New Age International Publishers, 8thEd, 2018.
3. S M Sze, Physics of Semiconductor Devices, Wiley, 3rd Ed, 2004.
4. Walter Fox Smith, “Experimental Physics”-Principles and Practice for the Laboratory, CRC Press, Taylor & Francis group, 1stEd, 2020.

Course Title	**Innovation & Entrepreneurship**				**Course Type**		**FC Integrated**	
Course Code	**B24ME0102**	**Credits**	**2**		**Class**		**I Semester**	
COURSE STTRUCTURE	LTP	Credits	Contact Hours	Work Load	Total Number of Classes Per Semester		Assessment Weightage	
	Lecture	1	1	1				
	Tutorial	-	-	-				
	Practical	1	2	2	Theory	Practical	CIE	SEE
	Total	**2**	**3**	**3**	**14**	**28**	**50%**	**50%**

COURSE OVERVIEW:

NEN Ignite is an entrepreneurship program based on experiential learning that aims to support startups' founders through a structured pathway from Idea Discovery to Pitch Deck. A 14 weeks, 36-42 hours of classroom/digital, highly experiential and practice based entrepreneurship training Course, by Wadhawani Foundation and will be delivered by WF facilitators / NEN Trained Entrepreneurship Faculty.

COURSE OBJECTIVES

The objectives of this course are to:

1. Discover an entrepreneurial opportunity
2. Articulate a compelling value proposition
3. Build a sustainable business model and business plan
4. Create and validate an MVP with potential customers
5. Select an appropriate Go-to-Market Strategy
6. Pitch the business idea to different stakeholders

COURSE OUTCOMES (CO'S)

On successful completion of this course; the student shall be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Identify the different aspects that can impact their business	**3,9,10,11**	**1**
CO2	Acquire in-depth knowledge about tools to build any business idea	**3,9,10,11**	**1**
CO3	Acquire in-depth knowledge about the different growth tools to grow their business.	**3,9,10,11**	**1**
CO4	Create a financial plan for their business	**3,9,10,11**	**1**

CO5	Create a pitch deck for their business and present it to different stakeholders	3,9,10,11	1

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

CO	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1			√			
CO2			√			
CO3			√			
CO4						√
CO5						√

COURSE ARTICULATION MATRIX

CO/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1			2					2	2	3	2	2		
CO2			2					2	2	3	2	2		
CO3			2					2	2	3	2	2		
CO4			2					2	2	3	2	2		
CO5			2					2	2	3	2	2		

COURSE CONTENTS

Introduction to Entrepreneurship: Entrepreneurs; entrepreneurial personality and intentions - characteristics, traits and behavioural; entrepreneurial challenges. Taking product or service ideas to creating value: Why should one choose to become an entrepreneur, Entrepreneurial mind-set, Intrapreneurship.

Orientation for WE Ignite program, Ice Breaking session, self-work Instructions and timelines Platform Demo Introduction to Ignite program flow and milestones , Introduction to Entrepreneurship and Human centred Approach to Design Thinking , Are you enterprising?. New generations of entrepreneurship viz. social entrepreneurship, Edupreneurship, Health entrepreneurship, Tourism entrepreneurship, Women entrepreneurship etc., Barriers to entrepreneurship, Creativity and entrepreneurship, Innovation and inventions, Skills of an entrepreneur, Decision making and Problem Solving

100 Rupee Venture: Debrief of Group Activity- Presentation and Sharing Learning Experience **Entrepreneurial Opportunities:** Opportunities. Discovery/ creation, Pattern identification and recognition for venture creation: prototype and exemplar model, reverse engineering. Problem Identification and Opportunity Discovery.

Entrepreneurial Process and Decision Making: Entrepreneurial ecosystem, Ideation, development and exploitation of opportunities; Negotiation, decision making process and approaches, Effectuation and Causation

Customer and Markets : Customer Discovery: Exploring Customer Personas & Market Estimation for your Ideas, Create a compelling value proposition & Competitive Advantage

Build your MVP : Building a MVP that customers Love

Crafting business models and Lean Start-ups: Introduction to business models; Creating value propositions-conventional industry logic, value innovation logic; customer focused innovation; building and analysing business models; Business model canvas, Introduction to lean start-ups, Business Pitching **Business Model:** Developing strong business models Create and present your Lean Canvas

Financial Feasibility: Introduction to Business plan and its components; Basics of Finance. **Institutional Support for Entrepreneurship:** Organization Assistance to an entrepreneur, New Ventures, Industrial Park (Meaning, features, & examples), Special Economic Zone (Meaning, features & examples), Financial assistance by different agencies, MSME Act Small Scale Industries, Carry on Business (COB) license, Environmental Clearance, National Small Industries Corporation (NSIC), e- tender process, Excise exemptions and concession, Exemption from income tax, The Small Industries Development Bank of India(SIDBI), Incentives for entrepreneurs

Go To market Strategy: Getting products to market: Channels & Strategies; Managing growth and Targeting Scale: Understand the Unit economics for your venture; Funding Strategy: Securing funding for your Startup and Preparing for pitch.

TEXTBOOKS:

1. K. Ramachandran, “Entrepreneurship Development”, Tata Mc. Graw Hill, 1st Edition, 2008
2. Sangeeta Sharma, “Entrepreneurship Development” PHI Publications, 1st Edition, 2022.

REFERENCE BOOKS:

1. Baringer and Ireland, “Entrepreneurship”, Pearson, 11th Edition, 2020.
2. Drucker F Peter: “Innovation and Entrepreneurship”, Heinemann, London, 1985.
3. Doanld F Kuratko and Richard M Hodgeth, “Entrepreneurship in the New Millennium”, India Edition - South-Western,
4. Robert D. Hisrich, “Entrepreneurship”, McGraw Hill, 11th Edition, 2020.
5. Donald F. Kuratko and Richard M. Hodgetts “Entrepreneurship: Theory, Process and Practice” South- Western, 6th Edition, 2003.
6. Thomas N. Duening, Robert D. Hisrich and Michael A. Lechter, “Technology Entrepreneurship: Taking Innovation to the Marketplace”, Academic Press, 2nd Edition, 2014.

JOURNALS/MAGAZINES

1. International Small Business Journal: https://journals.sagepub.com/home/isb
2. Journal of Development Entrepreneurship: https://www.worldscientific.com/worldscinet/jde

SWAYAM/NPTEL/MOOCs:

1. Entrepreneurship: https://nptel.ac.in/courses/110/106/110106141/

Course Title	**Communicative English**				**Course Type**		**FC**	
Course Code	**B24AH0103**	**Credit**	**1**		**Class**		**I Semester**	
Course Structure	**LTP**	**Credit**	**Contact Hours**	**Workload**	**Total Number of Classes Per Semester**		**Assessment Weightage**	
	Lecture	-	-	-				
	Tutorial	-	-	-	**Theory**	**Practical**	**CIE**	**SEE**
	Practical	**1**	**2**	**2**				
	Total	**1**	**2**	**2**	-	**28**	**25**	**25**

COURSE OVERVIEW

Today many companies are working on international projects where English is increasingly used by engineers across the world to communicate with all groups involved. This course is designed for entry-level Engineering and Technology curriculum enabling the students to learn, acquire, and apply for their learning and career. The course is aimed at providing effective skills for promoting communication skills through English. Hence in the present scenario Engineering English course is to be fully tailored to the specific needs of engineers. The outcome of this course contains a refined level of English proficiency by acquiring all four skills, listening, speaking, reading, and writing to prepare them for global readiness.

COURSE OBJECTIVE (S):

The objectives of this course are :

- To develop the communicative competence of learners
- To utilize language effectively in academic contexts
- To change various listening strategies to comprehend various types of audio materials like lectures, discussions, videos, etc.
- To build on students' interest in English language skills by engaging them in listening, speaking, and grammar learning activities that are relevant to authentic contexts.

COURSE OUTCOMES (S):

CO#	Course Outcomes	POs	PSOs
CO1	Analyse Listening, Comprehend and Correspond with others in various contexts	10, 11	
CO2	Apply proper communication modules to Speak legibly and fluently under various lifetime situations	10, 11	
CO3	Analyse the meaning and language while reading a variety of writings and technical texts	10, 11	
CO4	Demonstrate products and processes and explain their uses and purposes clearly and accurately	10, 11	
CO5	Make use of various communicative skills in precise and efficient ways in technological contexts	10, 11	
CO6	Create situational conversations and technical writing styles for interpersonal and effective communication	10, 11	

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1				√		
CO2			√			
CO3				√		

CO4		√				
CO5			√			
CO6						√

COURSE ARTICULATION MATRIX:

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1										2	2			
CO2										2	1			
CO3										2	2			
CO4										2	2			
CO5										2	1			
CO6										3	2			

Note:1-Low,2-Medium,3-High

COURSE CONTENTS

Fundamentals of Communication

Listening: Listening for general information-specific details- conversation: Introduction to classmates - Audio /video (formal & informal); Telephone conversation; Listening to voicemail & messages; Listening and filling a form. Listening to TV News and guest Lecturers, .

Speaking: Pronunciation Common Vocabulary – Technical Vocabulary – Answering Peer Questions – Conversation with Teacher, making telephone calls Introduction; Introducing a friend, Group Discussion

Reading: News magazines, Reading for unfamiliar words, Variety of News Items

Writing: Word formation – Auxiliary verbs – Modal Verbs – Sentence Types – Affirmative, Negative, Interrogative, Concord – Dialogue Writing, Paragraph writing

Narration and Summation

Listening: Listening to Peer Conversations – Brief Speeches – Listening for Specific Information – Recap of Speeches, Listening to podcasts, anecdotes/stories / event narration; documentaries and interviews with celebrities.

Speaking: Empathy, Telephonic enquiries official/formal enquiries, narrating personal experiences / events-Talking about current and temporary situations & permanent and regular situations - describing experiences and feelings- engaging in small talk- describing requirements and abilities, Debate.

Reading: Technical Essays – Identifying Sentence Types – Classifying the verb patterns

Writing: Tenses – Simple Present, Present Progressive, Present Perfect, Present Perfect Continuous – Voice – Active & Passive, email writing and Etiquesttes.

Description of A Process / Product

Listening: Listen to product and process descriptions; a classroom lecture; and advertisements about products. Digital Videos for World Information

Speaking: Picture description- describing locations in workplaces- Giving instruction to use the product- explaining uses and purposes- Presenting a product- describing shapes and sizes and weights- talking about quantities(large & small)-talking about precautions, Public Speaking (activity-based learning),Giving Feedback- Consructive Critisism.

Reading: Coherence, Development of Thoughts

Writing: Tenses – Simple Past, Past Progressive, Past Perfect, Past perfect continuous – Impersonal Passive-Narrating the past events, Permission Letter for Industrial Visit/Functions held

Classification and Recommendations

Listening: Listening to TED Talks; Listening to Dialects of English – British & American Regional Listening to achievers, eminent personalities

Speaking: Role Plays, Extempore, Responding to specific questions, Presentation Skills, Story Telling, Team building.

Reading: Texts on self-confidence, motivation, success path, Comprehensive passages, Reading for specific points

Writing: Tenses – Simple Future, Future progressive, Future Perfect, Future Perfect continuous – Definition – Phrases of Reason – Cause & Effect, Report writing

TEXT BOOKS:

1. Board of Editors. Using English A Course book for Undergraduate Engineers and Technologists. Orient Black Swan Limited, Hyderabad: 2015
2. Richards, C. Jack. Interchange Students' Book-2 New Delhi: CUP, 2015.

REFERENCE BOOKS:

1. Murphy, Raymond English Grammar in Use with Answers: Reference and Practice for Intermediate Students, Cambridge: CUP, 2004
2. Thomson, A.J. and Martinet, A.V. A Practical English Grammar, OUP, New Delhi: 1986
3. Anne Laws, ―Writing Skills, Orient Black Swan, Hyderabad, 2011
4. Green, David. Contemporary English Grammar Structures and Composition. MacMillan, 2010.
5. Thorpe, Edgar and Showick Thorpe. Basic Vocabulary. Pearson Education India, 2012.
6. Leech, Geoffrey and Jan Svartvik. A Communicative Grammar of English. Longman, 2003.
7. Rizvi, M. Ashraf. Effective Technical Communication. Tata McGraw-Hill, 2005.
8. Riordan, Daniel. Technical Communication. New Delhi: Cengage Publications, 2011.
9. Sen et al. Communication and Language Skills. Cambridge University Press, 2015.

Detailed Syllabus

Semester 2

Course Title	Probability and Statistics				Course Type		HC	
Course Code	B24AS0203	Credits	3		Class		II semester	
Course Structure	LTP	Credits	Contact	Work Load	Total Number of Cl asses Per Semester		Assessment Weightage	
	Lecture	3	3	3				
	Tutorial	-	-	-	Theory	Practical	CIE	SEE
	Practical	-	-	-				
	Total	**3**	**3**	**3**	**42**	**-**	**50%**	**50%**

COURSE OVERVIEW:

This course on Probability and Statistics for Computer Science introduces fundamental statistical methods and probability concepts with applications in computing. It covers curve fitting using least squares, correlation and regression techniques, and explores both discrete and continuous probability distributions such as Binomial, Poisson, Normal, and Exponential. Joint probability distributions, stochastic processes, and Markov chains are studied to understand system modeling and random processes. The course also introduces sampling theory, hypothesis testing, and goodness-of-fit tests. Special emphasis is given to applications of these methods in data science, machine learning, computer vision, computer graphics, and operating systems, enabling students to connect mathematical foundations with computational practices.

COURSE OBJECTIVE (S):

Student will be able to learn,

1. The concept of curve fitting and few statistical methods.
2. Fundamentals of probability- Random variables.
3. Joint probability and regarding stochastic process.
4. Concept of Test of Hypothesis and able to apply in the various fields of engineering.

COURSEOUTCOMES(COs)

After the completion of the course, the student will be able to:

CO#	Course Outcomes	POs	PSOs

CO1	Construct the data to a linear and non-linear equations using	1,2,4	1
CO2	Construct regression lines for the statistical data.	1,2,4	1
CO3	Define the concepts of probability density function and random variable for the various engineering problems.	1,2,4	1
CO4	Analyze the concepts of discrete and continuous probability distributions for various engineering problems.	1,2,4	1
CO5	Make use the concepts of stochastic processes and Markov chains in discrete and continuous transition state.	1,2,4	1
CO6	Apply the concepts of sampling theory to solve the various engineering problems.	1,2,4	1

BLOOM'S LEVELOF THECOURSE OUTCOMES

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1	√	√	√			
CO2	√	√	√			
CO3	√					
CO4	√	√	√	√		
CO5	√	√	√			
CO6	√	√	√			

COURSE ARTICULATIONMATRIX

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2		2								2		
CO2	3	3		2								2		
CO3	3	2		3								2		
CO4	3	3		2								2		
CO5	3	2		3								2		
CO6	3	3		2								2		

Note:1-**Low**,2-Medium,3-High

COURSE CONTENT THEORY

Contents
Curve Fitting: Curve fitting by the method of least squares and fitting of the curves of the form, $y = ax + b$, $y = ax^2 + bx + c$, $y = ae^{bx}$, $y = ax^b$ and $y=ab^x$. **Statistical Methods**: Correlation-Karl Pearson's coefficient of correlation- problems. Regression analysis- lines of regression, problems. Rank correlation. ** ***Application:*** *Curve fitting and statistics for data science*
Probability distributions: Random variables, Discrete and continuous probability distributions. Binomial, Poisson, normal distributions (only problems) and exponential (definition with one /two examples). ** ***Application:*** *Probability distribution in machine learning, Computer vision: object recognition and image segmentation* *Computer graphics: behaviour of light and other physical phenomena in computer graphics*
Joint Probability distribution: Joint Probability distribution for two discrete random variables, expectation, covariance, correlation coefficient. Stochastic processes- Stochastic processes, probability vector, stochastic matrices, fixed points, regular stochastic matrices, Markov chains, higher transition probability-simple problems. ** **Application:** *Stochastic processes and Markov processes in Operating System*
Sampling theory: Sampling, Sampling distributions, standard error, test of hypothesis for means and proportions, confidence limits for means, student's t-distribution. Chi-square distribution as a test of goodness of fit. ** ***Application***: **S***ampling process in computer graphics, sampling theory in machine learning*

** *Application: (i) Additional information providing to students only for knowledge.*

(ii) Major part of assignments questions chosen from applications.

(iii) Students' presentations/seminars topics chosen from applications.

TEXTBOOKS:

1. B.S. Grewal, "Higher Engineering Mathematics",43nd edition Khanna Publishers, 2015.
2. ErwinKreyszig, "Advanced Engineering Mathematics",9thedition Wiley Publications, 2013.
3. Seymour Lipschutz, John J. Schiller., "Schaum's Outline of Introduction to Probability and Statistics", McGraw Hill Professional, 1998.

REFERENCEBOOKS:

1. B.V. Ramana, "Higher Engineering Mathematics", 19th Reprint edition, Tata McGraw Hill Publications, 2013.
2. R.K.Jain and S.R.K.Iyengar, "Advanced Engineering Mathematics",4th edition,Narosa Publishing House, 2016.
3. V.Sundarapandian, "Probability, Statistics and Queueing theory", PHI Learning, 2009
4. Dr. B. Krishna gandhi, Dr. T.K.V. Iyengar, Dr. M.V.S.S.N. Prasad&S. Ranganatham.,"Probability and Statistics"S. Chand Publishing, 2015.
5. J. K. Sharma "Operations Research theory and applications", 5th edition, Macmillan publishers, ,2013.

JOURNALS/MAGAZINES

1. https://www.hindawi.com/journals/jps/
2. https://www.journals.elsevier.com/statistics-and-probability-letters

3. http://www.isoss.net/japs/

SWAYAM/NPTEL/MOOCs:

1. https://www.coursera.org/browse/data-science/probability-and-statistics
2. https://nptel.ac.in/courses/111/105/111105041/
3. https://onlinecourses.swayam2.ac.in/cec20_ma01/preview

SELF-LEARNING EXERCISES:

1. Curve fitting techniques for application-oriented problems; Regression analysis for bivariate data.
2. Exploration of probability distributions: Geometric and Gamma distributions; Joint probability distribution of continuous random variables.
3. Sampling analysis of real-time problems with applications to computer science, including data mining, pattern classification, and related domains.

<table>
<tr><td>Course Title</td><td colspan="4">Chemical Technology for Computing</td><td colspan="2">Course Type</td><td colspan="2">HC</td></tr>
<tr><td>Course Code</td><td>B25AS0105</td><td>Credits</td><td colspan="2">3</td><td colspan="2">Class</td><td colspan="2">II semester</td></tr>
<tr><td rowspan="5">Course Structure</td><td>LTP</td><td>Credits</td><td>Contact</td><td>Work Load</td><td colspan="2" rowspan="2">Total Number of Classes Per Semester</td><td colspan="2" rowspan="2">Assessment Weightage</td></tr>
<tr><td>Lecture</td><td>2</td><td>2</td><td>2</td></tr>
<tr><td>Tutorial</td><td>1</td><td>2</td><td>2</td><td rowspan="2">Theory</td><td rowspan="2">Practical</td><td rowspan="2">CIE</td><td rowspan="2">SEE</td></tr>
<tr><td>Practical</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>Total</td><td>3</td><td>4</td><td>4</td><td>32</td><td>32</td><td>50%</td><td>50%</td></tr>
</table>

COURSE OVERVIEW:

Chemical Technology for Computing integrates highly pertinent subjects that are suitable for engineering students, enlightening them about the significance of several facets of fundamental science in engineering. Chemical Technology for Computing includes the study of instrumental technology analysis methods, technologies for clean and green energy storage and conversion devices, and novel materials for electronic devices. Additionally, the course emphasizes the study of the chemical properties and uses of functional materials in several engineering fields. This field of science is highly interdisciplinary and provides an opportunity for students to enhance their understanding of engineering principles concerning sustainable energy, conversion, and storage devices, semiconducting materials, and polymeric materials. These areas have become especially interesting for engineering research. The topic encompasses a wide range of engineering materials, their characteristics, and their use in the engineering domain.

COURSE OBJECTIVES:

The objectives of the course are to:

1. Understand the electronic transitions in materials, instrumental methods of analysis in technology using various instruments.
2. Illustrate renewable energy storage and conversion device technologies for batteries, super capacitors of operating mechanisms and applications.

3. Acquire a comprehensive understanding of advanced materials used in electronic devices, with a specific emphasis on the significance of semiconducting materials and the principles of metal finishing in industrial procedures.
4. Identify efficient polymeric materials, liquid crystals in display technologies and electronic applications.

COURSE OUTCOMES (COs):

On successful completion of this course, the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Demonstrate the principles and uses of clean-green energy storage and conversion device technologies.	1,2,3,5,8	1
CO2	Interpret the instrumental methods of analysis in technology, electronic transitions in materials and the detection of samples using various sensors.	1,2,3,6,8	1
CO3	Identify the significance of modern materials in electronic devices and the theory of electroplating in industrial techniques.	1,2,3,6,8	1
CO4	Analyze the importance of polymeric materials and Nano materials for various applications.	1,2,3,8	1
CO5	Illustrate the fundamental principles of electrode systems, batteries, and super capacitors.	1,2,3,6,8	1
CO6	Explain the practical uses of advanced materials in engineering and technology.	1,2,3,6,8	1

BLOOM’S LEVEL OF THE COURSE OUTCOMES:

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1			√			
CO2			√			
CO3				√		
CO4				√		
CO5			√			
CO6				√		

COURSE ARTICULATION MATRIX:

COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	2	2		1		3				1		
CO2	2	2	3			2		2				1		
CO3	2	1	1			1		3				1		
CO4	3	2	2			1		2				1		
CO5	3	2	1			2		1				1		
CO6	2	2	1			1		1				1		

Note:1-Low; 2-Medium; 3-High

Contents
Clean Energy Storage and Conversion Device Technologies **Electrode System:** Types of electrodes, applications of electrodes in agriculture and environment. **Energy storage devices:** Battery components and characteristics: Primary battery (dry cell), secondary battery (lead-acid, Li-ion battery) **Hands-on-practice:** Assembly and Fabrication of Galvanic Cell. **Case study:** Choice of battery materials for e-vehicles **Supercapacitors:** The electrochemical double Layer-Helmholtz model, charge accumulation mechanisms in EDLC, pseudo and hybrid capacitors with applications. **Photovoltaic cell:** Principle, working, advantages and applications. **Hands-on-practice:** Fabrication of dye sensitized solar cell **Sensors:** Introduction, types, principles, and applications of thermal, piezoelectric, electrochemical, bio, and gas sensors. **Instrumentation:** Principles, instrumentation, and applications of UV-visible spectrophotometry, flame photometry, potentiometry, and conductometry. **Hands-on-practice**: Quantitative analysis of Biomolecules/sodium/ potassium/lithium and iron.

Electroless plating: Principle of electro less plating, manufacture of Printed circuit board using electroless plating of copper and applications.
Hands-on-practice: Electro plating and electro less plating of copper.
Case study: Materials for printed circuit boards.
Polymeric materials: Introduction, biodegradable and biocompatible polymeric materials. Conducting polymers-electron transport mechanism, and applications of polyacetylene and polyaniline (PANI).
Hands-on-practice: Thin film Fabrication of a conducting polymer for device applications by a) electro polymerization b) chemical oxidative method
Liquid crystals: Classification, and applications in electronic display systems.

Nanomaterials: Introduction, classification, and quantum confinement. Size-dependent properties - surface area, electrical properties, optical properties, magnetic properties, and thermal properties.
Hands-on-practice: Preparation of semiconducting nanomaterials.
Smart materials for signal conversion: Solvetochromism, Mechanochromism, Piezoelectric materials, shape memory alloys, electro active materials and photochromic materials.
Hands-on-practice: Study of optical band gap of semiconducting materials by UV-Visible spectrophotometer
Case study: Recent advances in nanotechnology for flexible electronic devices**.**

REFERENCE BOOKS:

1. Mr. M. Sivasubramanian, Introduction to Sensors and Transducers, 2022, Notion Press
2. V.R. Gowariker, N.V. Viswanathan & J. Sreedhar., "Polymer Science", Wiley-Eastern Ltd. Callister W.D., "Materials Science and Engineering", John Wiley & Sons, 10th edition, 2013.
3. B. M Chandrashekar and B.C. Basavaraju, ''Engineering Chemistry'', Basuchandras Publishing 3rd edition 2016.
4. Ahmed Barhoum, Fundamentals of Sensor Technology: Principles and Novel Designs, Woodhead Pub Ltd, 2023
5. O.V. Roussak and H. D. Gesser , Applied Chemistry: A textbook for Engineers and Technologists, Springer New York, NewYork, 2nd edition, 2013.
6. B. Bhardwaj ''Applied Chemistry'', S.K. Kataria & Sons, 2nd edition, 2024.
7. T. Pradeep, Textbook of Nanoscience and Nanotechnology, 2012 McGraw Hill Education (India) Private Limited.
8. Samuel Glasstone, An Introduction to Electrochemistry, 2006, East-West Press (Pvt.) Ltd.

Fast Learners:

1. Average molecular weight determination by viscosity method.
2. Industrial Wastewater treatment by Photocatalysis.
3. Preparation of Polystyrene and Phenol - Formaldehyde or Urea - Formaldehyde Resin.
4. Membrane preparation for water purification.

Self-Learning Component: Chemical Structure drawing (2D &3D), energy minimization by using ISIS draw, Avogadro software, Auto Dock software, and Various types of plots using Origin software.

TEXTBOOKS:

1. R.V. Gadag and Nithyanandashetty N, "Engineering Chemistry" Ik International Publishing house, 3rd edition, 2014.
2. B. Siva Shankar, "Engineering Chemistry", Tata Mc Graw Hill Publishing Limited, 3rd Edition, 2015.
3. S. S. Dara, Mukkanti, "Text of Engineering Chemistry", S. Chand & Co, New Delhi, 12th Edition, 2006.
4. Bhatnagar M.S., A Textbook of Polymers, 2010, S Chand & Company
5. B.S. Jai Prakash, R.Venugopal, Sivakumaraiah & Pushpa Iyengar., "Chemistry for Engineering Students", Subhash Publications, Bangalore.

6. R. P. Mani, K. N. Mishra, "Chemistry of Engineering Materials", Cengage Learning, 3rd Edition, 2015.
7. P. C. Jain, Monica Jain, "Engineering Chemistry", Dhanpat Rai Publishing Company, 17th Edition, 2015.
8. D. Singh, B. Deshwal, S. Vats, "Comprehensive Engineering Chemistry", Dreamtech Press, 2020.

Additional Links:

1. https://www.youtube.com/watch?v=JlZhHhpVRrI
2. https://www.youtube.com/watch?v=xAS4NS9RuI4
3. https://www.youtube.com/watch?v=KvdlWobaBkM

<table>
<tr><td>Course Title</td><td colspan="4">Advanced C programming with Copilot</td><td colspan="2">Course Type</td><td colspan="2">HC</td></tr>
<tr><td>Course Code</td><td>B25CI0201</td><td>Credits</td><td colspan="2">2</td><td colspan="2">Class</td><td colspan="2">II Semester</td></tr>
<tr><td rowspan="5">COURSE STRUCTURE</td><td>LTP</td><td>Credits</td><td>Contact Hours</td><td>Workload</td><td colspan="2" rowspan="3">Total number of classes per semester</td><td colspan="2" rowspan="3">Assessment Weightage</td></tr>
<tr><td>Lecture</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>Tutorial</td><td>1</td><td>2</td><td>2</td></tr>
<tr><td>Practice</td><td>-</td><td>-</td><td>-</td><td>Theory</td><td>Practical</td><td>CIE</td><td>SEE</td></tr>
<tr><td>Total</td><td>2</td><td>3</td><td>3</td><td>42</td><td>-</td><td>50%</td><td>50%</td></tr>
</table>

COURSE OVERVIEW:
This course is designed to equip students with the skills necessary to leverage AI-powered tools in modern software development. With an emphasis on hands-on coding and practical project development, the course introduces students to key technologies and best practices that enhance programming productivity using AI copilots along with covering essential C programming concepts. Students will gain practical experience by applying these skills to solve real-world problems. Additionally, the course introduces UNIX operating systems and shell scripting, focusing on basic commands and automation tasks. By the end, students will have a solid understanding of developing complex programs using Copilot, C programming and UNIX scripting, preparing them for more advanced programming challenges.

COURSE OBJECTIVE(S):
The objectives of this course are to:

- **Leverage AI-powered tools** like GitHub Copilot to streamline coding and boost productivity.
- **Master shell scripting** by navigating the UNIX file system and writing efficient Bash scripts for automation.
- **Enhance C programming skills** through a deep understanding of memory management, pointers, and dynamic memory allocation.
- **Implement file input/output operations** effectively using C standard library functions for file handling and error checking.
- **Understand and implement key data structures** such as stacks, queues, and linked lists to solve real-world problems efficiently.
- **Build full-stack applications** using Streamlit, FastAPI, and MongoDB for dynamic web development.
- **Collaborate on coding projects** using GitHub and GitHub Copilot to improve teamwork and version control practices.
- **Apply critical problem-solving skills** to complete hands-on projects and use cases that reinforce the learning of core programming concepts.

COURSE OUTCOMES (COs):

After the completion of the course, the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Develop modular and reusable code using functions (call by value and call by reference), structures, arrays and arrays of structures using illustrative programs and test-driven development.	1-5,9, 11, 12	1-3
CO2	Develop modular and reusable code using functions (call by value and call by reference), structures, arrays and arrays of structures using GitHub copilot using illustrative programs and test-driven development.	1-3, 5	
CO3	Develop modular and reusable code using pointers and dynamic memory allocation using illustrative programs and test-driven development.	1-3	
CO4	Develop modular and reusable code using pointers and dynamic memory allocation using GitHub copilot using illustrative programs and test-driven development.	1-3	2
CO5	Develop a simple application with strings and file handling in C using illustrative programs and test-driven development	1-3, 5, 9-12	1-3
CO6	Develop a simple application with strings and file handling in C using GitHub copilot using illustrative programs and test-driven development	1-5,9, 11	1-3

BLOOM'S LEVELOF THECOURSE OUTCOMES

CO#	Bloom's Level					
	Remember (L1)	**Understand (L2)**	**Apply (L3)**	**Analyse (L4)**	**Evaluate (L5)**	**Create (L6)**
CO1			✓			
CO2			✓			
CO3		✓				
CO4			✓			
CO5			✓			
CO6				✓		

COURSE ARTICULATIONMATRIX:

CO# / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	3				3		3	3	2	3
CO2	2	2	2		3									
CO3	2	2	2											
CO4	2	2	3										2	

CO5	3	3	3		3				3	3	3	3	2	3
CO6	3	3	3	3	3				3		3	3	2	3

Note: 1-Low, 2-Medium, 3-High

COURSE CONTENT THEORY

Content
1. **Review of C Fundamentals:** Quick review of basic C programming concepts. Control and looping statements, arrays, functions (call by value and call by reference) 2. **Introduction to AI-Powered Code Assistants:** Introduction to AI copilot, Integration with IDEs, Prompt Engineering Basics, Iterative refinement of prompts, Understanding Copilot-Generated Code (Code analysis and explanation, Identifying logic flow and dependencies) 3. **Structures:** Introduction to Structures, Defining and Using Structures, Arrays of structures 4. **Pointers:** Understanding pointers, Dynamic Memory Management using malloc and free, pointers and arrays, pointers and strings, Pointers to Structures, singly linked lists as an application of pointers and self-referential structures 5. **File Handling:** File handling functions for various operations: Opening a File: fopen(), Reading from a File: fread(), fscanf(), Writing to a File: fwrite(), fprintf(), Closing a File: fclose(), Error Checking: feof(), Different modes of operations in these (r,w,a, binary)

TEXT BOOKS:

1. B.W. Kernighan & D.M. Ritchie, "C Programming Language", 2nd Edition, PRENTICE HALL SOFTWARE SERIES, 2005.
2. Herbert Schildt, “C: The Complete Reference”, 4th edition, TATA McGraw Hill, 2000.
3. B.S. Anami, S.A. Angadi and S. S. Manvi, “Computer Concepts and C Programming: A Holistic Approach”, second edition, PHI,2008.
4. Yashavant Kanetkar, “Let us C” , BPB , 19th edition, 2022
5. The Unix Programming Environment, by Brian Kernighan and Rob Pike, 1st Edition, Pearson Education India, 2015
6. Unix: Concepts and Applications by Sumitabha Das, 4th Edition, McGraw Hill Education

REFERENCES:

1. https://github.com/features/copilot
2. Balaguruswamy,” Programming in ANSI C”, 4th edition, TATA MCGRAW Hill, 2008.
3. Donald Hearn, Pauline Baker,“ Computer Graphics C Version”, second edition, Pearson Education, 2004.
4. Jon Bentley, “Programming Pearls”, 2nd edition, PEARSON,2002.
5. “LangChain for LLM Application Development” by DeepLearning.AI
6. “Multi AI Agent Systems with crewAI” by DeepLearning.AI
7. “Agentic AI and AI Agents: A Primer for Leaders” on Coursera

JOURNALS/MAGAZINES:

1. https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=6294 (IEEE Journal/Magazine on IT Professional)
2. https://ieeexplore.ieee.org/document/1267572 (IEEE Computing in Science and Engineering)

SWAYAMNPTEL/MOOCs

1. https://online courses.nptel.ac.in/noc20_cs06/preview (Problem Solving through Programming in C)
2. https://www.edx.org/course/c-programming-getting-started (C Programming Getting started)
3. https://www.coursera.org/specializations/c-programming (Introduction to C programming)

Course Title	**Electronics and Digital Logic**				**Course Type**		**HC**	
Course Code	**B25EE0101**	**Credits**	**3**		**Class**		**I/II semester**	
Course Structure	LTP	Credits	Contact	Work Load	Total Number of Classes Per Semester		Assessment Weightage	
	Lecture	3	3	3				
	Tutorial	-	-	-	Theory	Practical	CIE	SEE
	Practical	-	-	-				
	Total	**3**	**3**	**3**	**42**	**-**	**50%**	**50%**

PREREQUISITES:

1. Basic electrical concepts: Voltage, current, resistance, Ohm's Law
2. DC circuit analysis: Series/parallel circuits, KVL & KCL
3. Fundamentals of semiconductors: p-type, n-type materials
4. Number systems: Binary, decimal, conversions, binary arithmetic.
5. Basics of Boolean algebra and logic gates

COURSE OVERVIEW:

This course introduces the fundamental concepts of semiconductor devices, digital systems, and logic design. It starts with the quantitative theory of P-N junction diodes, exploring their characteristics and applications such as rectifiers and clippers. The course addresses the concepts, principles and techniques of designing digital systems. The course teaches the fundamentals of digital systems applying the logic design and development techniques. Through theory and practical design exercises, students gain a solid foundation in electronic devices and digital circuit design.

COURSE OBJECTIVE(S):

The objectives of this course are to:

1. Understand the physics and quantitative theory behind P-N junction diodes and their behaviour as rectifiers and limiters.
2. Analyze and design various diode-based rectifier circuits and clipping/clamping circuits.
3. Apply Boolean algebra principles to simplify logic functions and implement logic gates and circuits.
4. Perform gate-level minimization using Karnaugh maps and implement combinational logic circuits like adders, subtractors, multiplexers, de-multiplexers and comparator.
5. Understand and design ripple and synchronous sequential circuits including flip-flops, registers, and counters.

COURSE OUTCOMES (COs):

After the completion of the course, the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Explain the working principle and characteristics of P-N junction diodes and their applications.	PO1, PO2	2
CO2	Apply the principles of rectifier, clipping, and clamping circuits to signal waveforms.	PO1, PO2, PO3	1
CO3	Boolean algebra laws and theorems to simplify and manipulate logical expressions.	PO1, PO2, PO3, PO4, PO5	2
CO4	Interpret Boolean functions and express them in canonical and standard forms.	PO1, PO2, PO3, PO4, PO5	3
CO5	Simplify Boolean expressions and design digital circuits using basic and universal logic gates. Minimize logic functions using Karnaugh maps	PO1, PO2, PO3, PO4	1
CO6	Interpret the behavior of combinational and sequential logic circuits through examples of adders, subtractors, multiplexers, demultiplexers, and comparators.	PO1, PO2, PO3, PO4	1

BLOOM'S LEVELOF THECOURSE OUTCOMES

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluat (L5)	Create (L6)
CO1		✓				
CO2			✓			
CO3			✓			
CO4		✓				
CO5			✓			
CO6		✓				

Course Articulation Matrix:

CO# / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PS O1	PS O2	PS O3
CO1	3	2	1	1	1							3	1	3
CO2	3	2	1	1	1							3	1	3
CO3	3	2	1	1	1							3	1	3
CO4	3	2	1	1	1							3	1	3
CO5	3	2	2	1	1							3	2	3
CO6	3	2	2	1	1							3	2	3

Note: 1-Low, 2-Medium, 3-High

COURSE CONTENT THEORY

COURSE CONTENTS
P-N Junction diode: Quantative Theory of P-N Junction, P-N Junction as a diode, diode equation, V-I characteristic, ideal versus practical –resistance levels (static and dynamic) **Rectifiers And Limiters:** P-N Junction as a rectifier, Half wave rectifier, full wave rectifier, Bridge rectifier, Positive Clipping, Negative Clipping and clamping circuits using diodes. **Unit 2** **Boolean Algebra And Logic Gates:** Number System Conversions, digital logic gates other logic operations, axiomatic definition of Boolean algebra, basic theorems and properties of Boolean algebra, Boolean functions, canonical and standard forms. **Unit 3** **Gate Level Minimization:** Maxterm, Minterm, K-map method, two, three, and four-variable map, POS and SOP simplification, don't-care conditions, NAND and NOR implementation. **Unit 4** **Combinational Circuits:** Half/Full Adder & Subtractor, Ripple carry Adder, Carry-Look Ahead adder, Multiplexers, De-multiplexer, Encoder, Decoder, Comparators (1 and 2) **Sequential Logic**: Introduction to sequential circuits, Latches- SR, Flipflops: D, T, SR, JK, The master-slave flip-flops. Application & Case Studies

TEXTBOOKS

1. Jacob Millman, Christos Halkias, Chetan D Parikh," Millman's Integrated Electronics – Analog and Digital Circuits and Systems", 4th Edition, Tata McGraw Hill, 2015.
2. Anil K Maini, Varsha Agarwal," Electronic Devices and Circuits", 2nd edition, Wiley, 2019.
3. Donald P Leach, Albert Paul Malvino&GoutamSaha," Digital Principles and Applications", 8th Edition, Tata McGraw Hill, 2014.

REFERENCES

1. R D Sudhaker Samuel, "Illustrative Approach to Logic Design", Sanguine-Pearson, 2010.
2. Charles H. Roth," Fundamentals of Logic Design", Jr., 7th Edition, Cengage Learning, 2024.
3. Ronald J. Tocci, Neal S. Widmer, Gregory L. Moss," Digital Systems Principles and Applications", 12th Edition, Pearson Education, 2017.
4. M Morris Mano," Digital Logic and Computer Design", 10th Edition, Pearson Education, 2008.
5. Jacob Millman, Christos Halkias, "Analog and Digital Circuits and Systems", 2nd Edition, Tata McGraw Hill, 2010
6. R. D. Sudhaker Samuel," Electronic Circuits", Sanguine-Pearson, 2010

CERTIFICATION COURSES (OPTIONAL / RECOMMENDED)

1. NPTEL / SWAYAM Courses:
   i. *Digital Circuits*
   ii. *Basic Electronics*
   iii. *Switching Circuits and Logic Design*
2. Coursera / edX:

i. *Introduction to Digital Systems*
ii. *Semiconductor Devices and Circuits*

3. Texas Instruments / Intel FPGA: Online logic design workshops

Tools to be Used

1. **Digital Logic Simulator** (e.g., Logisim, Digital Works)
2. **Multisim / Proteus** (for circuit simulation)
3. **Breadboard and Electronic Components** (for hands-on diode and rectifier circuits)
4. **Verilog / VHDL** (for advanced digital logic implementation – optional)
5. **Online Truth Table & K-Map Solvers**

<table>
<tr><td>Course Title</td><td colspan="4">Python for Data Science</td><td>Course Type</td><td colspan="2">HC</td></tr>
<tr><td>Course Code</td><td>B25CS0101</td><td>Credits</td><td colspan="2">2</td><td>Class</td><td colspan="2">I Sem</td></tr>
<tr><td rowspan="5">Course Structure</td><td>LTP</td><td>Credits</td><td>Contact hours</td><td>Workload</td><td rowspan="2">Total Number of Classes Per Semester</td><td rowspan="2" colspan="2">Assessment in Weightage</td></tr>
<tr><td>Lecture</td><td>2</td><td>2</td><td>2</td></tr>
<tr><td>Tutorial</td><td>-</td><td>-</td><td>-</td><td rowspan="2">Theory</td><td rowspan="2">IA</td><td rowspan="2">SEE</td></tr>
<tr><td>Practical</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>Total</td><td>2</td><td>2</td><td>2</td><td>28</td><td>50</td><td>50</td></tr>
</table>

Pre-requisite

NA

Overview of Course

This course covers the basics of programming, data handling, visualization techniques, and real-world data science applications. Students will develop practical skills through hands-on lab sessions using industry-standard tools and libraries, preparing them for advanced courses and interdisciplinary projects in the field of Data Science

Course Objectives

- Understand the syntax, semantics, and core concepts of Python programming.
- Acquire foundational skills in data structures and data manipulation using Python.
- Explore essential libraries for data analysis, such as NumPy and Pandas.
- Develop proficiency in data visualization using Matplotlib and Seaborn.
- Apply Python to solve basic data science problems and build simple analytical models.

Course Outcomes

Upon completion, students will be able to:

CO #	Course Outcome Statement
CO1	**Demonstrate**, **interpret**, and **apply** basic Python programming constructs including syntax, variables, and control structures. *(Unit I)*
CO2	**Develop**, **manipulate**, and **implement** programs using core Python data structures (lists, tuples, sets, dictionaries). *(Unit II)*
CO3	**Perform**, **analyze**, and **clean** datasets using Pandas and NumPy libraries for data wrangling. *(Unit II)*
CO4	**Visualize**, **customize**, and **evaluate** data using Matplotlib and Seaborn libraries. *(Unit III)*
CO5	**Analyze**, **extract**, and **present** insights from structured datasets using Python libraries. *(Unit IV)*
CO6	**Design**, **execute**, and **demonstrate** mini-projects addressing real-world data science problems. *(Unit IV)*

BLOOM'S LEVELOF THECOURSE OUTCOMES

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO-1		√				
CO-2			√			
CO-3				√		
CO-4				√		
CO-5					√	
CO-6					√	

Course Articulation Matrix:

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	2	1	3				1	2	1	3		
CO2	3	2	3	2	3				2	2	1	3		
CO3	3	3	2	3	3				1	2	1	3	2	1
CO4	2	2	2	2	3				2	3	2	3	2	1
CO5	2	3	2	3	3				2	3	3	3	3	1
CO6	2	2	3	3	3	1	1	1	3	3	3	3	3	1

Contents
Python Basics Introduction to Python: History and features, Installation and setup, Python Syntax, Variables, Data Types, Operators and Expressions, Input/Output Operations, Control Structures: Conditional statements, Loops, Functions, Modules, and Packages, Error Handling and Debugging **Working with Data** Data Structures: Lists, Tuples, Sets, Dictionaries, File Handling: Reading and writing files, Introduction to NumPy: Arrays and array operations, Data Manipulation with Pandas: Series and Data Frames, Data Cleaning: Handling missing values, Data type conversion **Data Visualization and Libraries** Introduction to Data Visualization, Plotting with Matplotlib: Line, Bar, Scatter, Pie charts, Advanced Visualization with Seaborn, Customizing Plots: Labels, legends, styles, Introduction to Exploratory Data Analysis (EDA) **Mini Projects and Data Science Applications** Framing Data Science Problems, Mini Projects: Real-world Data Analysis: Weather, sports, social media datasets, Introduction to Machine Learning Concepts, Using Scikit-learn for Simple Predictive Models, Best Practices in Data Science Projects

Textbooks:

1.Jake VanderPlas, *Python Data Science Handbook*, O'Reilly, 2016
2.Wes McKinney, *Python for Data Analysis*, O'Reilly, 2017
3.Joel Grus, *Data Science from Scratch: First Principles with Python*, O'Reilly, 2019

References

1. Official Python Documentation (python.org)
2.NumPy and Pandas Documentation (numpy.org, pandas.pydata.org)
3.Coursera: "Python for Everybody" by University of Michigan
4.edX: "Introduction to Python for Data Science" by Microsoft
5.YouTube: Corey Schafer's Python Tutorials, freeCodeCamp Python Course
6.DataCamp, Kaggle Python Micro-courses

Certification courses

1. Coursera: "Applied Data Science with Python" by University of Michigan
2. edX: "Data Science Essentials" by Microsoft
3. NPTEL: "Python for Data Science"
4. Udacity: "Introduction to Data Science with Python"

List of Lab Activities

1. Writing basic Python programs (Hello World, arithmetic operations)
2. Implementing control structures and functions in Python
3. Manipulating lists, tuples, sets, and dictionaries
4. Reading and writing data from text and CSV files
5. Performing array operations using NumPy
6. Data manipulation and cleaning using Pandas

7. Creating various plots with Matplotlib and Seaborn
8. Exploratory Data Analysis on sample datasets
9. Building a simple predictive model with scikit-learn
10. Mini project: Analyzing a real-world dataset end-to-end

Tools to be Used

1. Jupyter Notebook
2. Google Colab
3. Anaconda Distribution
4. Python IDEs: VS Code, PyCharm, Spyder
5. Python 3.x (latest stable version)
6. Libraries: NumPy, Pandas, Matplotlib, Seaborn, scikit-learn

<table>
<tr><td>Course Title</td><td colspan="4">Fundamentals and Applications of Civil Engineering</td><td colspan="2">Course Type</td><td colspan="2">HC</td></tr>
<tr><td>Course Code</td><td>B24ED0102</td><td>Credits</td><td colspan="2">2</td><td colspan="2">Class</td><td colspan="2">I Sem</td></tr>
<tr><td rowspan="5">Course Structure</td><td>LTP</td><td>Credits</td><td>Contact hours</td><td>Workload</td><td colspan="2" rowspan="2">Total Number of Classes per semester</td><td colspan="2" rowspan="2">Assessment in Weightage</td></tr>
<tr><td>Lecture</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>Tutorial</td><td>1</td><td>2</td><td>2</td><td rowspan="2">Theory</td><td rowspan="2">Practical</td><td rowspan="2">CIE</td><td rowspan="2">SEE</td></tr>
<tr><td>Practical</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>Total</td><td>2</td><td>3</td><td>3</td><td>42</td><td>-</td><td>50</td><td>50</td></tr>
</table>

COURSE OVERVIEW

This course, serves as an introductory exploration into the diverse realms of civil engineering, emphasizing the integration of software tools and applications. It covers fundamental fields of civil engineering, including infrastructure development, construction materials, traffic management, Geo Informatics and smart city solutions. The course aims to bridge traditional civil engineering practices with modern techniques, providing students with the knowledge and skills necessary to address contemporary engineering challenges.

COURSE OBJECTIVES

The objective of this course are to

1. Know the various fields and scope of Civil Engineering and the socio-economic impact of different infrastructure types.
2. To develop the ability to analyze forces, support reactions, and loads in engineering mechanics, focusing on both concurrent and non-concurrent force systems.
3. To analyze and visualize the load distribution, shear force, bending moments and deflections of statically determinate beams using tools.
4. To understand the principles of transportation engineering, classification of roads, pavements, application of Intelligent Transportation Systems (ITS) and smart cities.

5. Use of geo-informatics principles, including remote sensing, GIS, and image processing, and apply these concepts in practical scenarios through case studies.
6. Create cross-sectional views of different road types and understand their practical applications.

COURSE OUTCOMES

After completion of the course students will be able to:

CO1: Understand the scope and fields of Civil Engineering, types of infrastructure, and their socio-economic impact.

CO2: Analyze forces, support reactions, and loads in engineering mechanics.

CO3: Analyse and visualize the load distribution, shear force, bending moments and deflections of statically determinate beams using tools.

CO4: Understand transportation engineering principles, including road classification, ITS and smart cities.

CO5: Utilize geo-informatics principles for remote sensing, GIS, and image Processing.

CO6: Apply software tools and create road cross-sections and understand their practical applications.

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

	Remember (L1)	**Understand (L2)**	**Apply (L3)**	**Analyze (L4)**	**Evaluate (L5)**	**Create (L6)**
CO1	✓	✓				
CO2		✓	✓	✓		
CO3		✓	✓			
CO4		✓	✓			
CO5		✓	✓	✓		
CO6		✓	✓	✓		

COURSE ARTICULATION MATRIX

CO#	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**PSO3**	**PSO4**
CO1	3	3				3					3	3			
CO2	3	3		2				3	3		3	3	3	1	
CO3	3	3		3		2		3	3		3	3	3	1	
CO4	3	3				3			3		3	3	3	1	
CO5	3	3		3		3					3	3	3		1
CO6	3	3									3	3	3		1

Contents

Introduction to Civil Engineering - Scope of different fields of Civil Engineering. Types of infrastructure, Importance of Infrastructural Development, Effect of the infrastructural facilities and social-economic development of a country. Construction materials.

Engineering Mechanics: Forces and Classification, analysis of concurrent force systems and non-concurrent force systems-resultants. Types of loads, supports and beams. Numerical problems on beams: Calculation of support reactions with point load, uniformly distributed load and moments.
Load applications and analysis: To visualize the load distribution, shear force, bending moments and deflections of statically determinate beams using Staad Pro/MS Excel.

Transportation Engineering: Classification of Roads and their functions, Flexible and Rigid Pavements. Road safety and management, Causes - Collection of data for traffic studies, Traffic management techniques, ITS and its application. **Smart cities:** Smart energy resources, Smart disposal of solid waste, smart road construction.
Create cross-sectional views of different road types using AutoCAD software

Geo Informatics: Principles of remote sensing - Satellites and Sensors, Introduction to GIS principles, Global positioning system, Introduction to image processing and its application. Case studies.

TEXT BOOKS

1. Elements of Civil Engineering Jagadeesh, T. R., & Jayaram, M. A. Sapna Book House. ISBN: 978-8120341675
2. Engineering Mechanics. Khurmi, R. S. S. Chand & Company Ltd. ISBN: 978-8121901014
3. Environmental Engineering. Rao, P. V. (2002)Prentice-Hall of India Pvt. Ltd. ISBN: 978-8123919352
4. Remote Sensing and GIS Bhatta, B. Oxford University Press. ISBN: 978-0198072393
5. Traffic Engineering and Transport Planning. Kadiyali, L. R. Khanna Publishers. ISBN: 978-8185596780
6. Building Construction. Punmia, B. C., Jain, A. K., & Jain, A. K. Laxmi Publications. ISBN: 978-8131804289

REFERENCE BOOKS

2. Structural Health Monitoring. Balageas, D., Fritzen, C. P., & Güemes, A. John Wiley & Sons. ISBN: 978-0470394381
3. Principles of Geotechnical Engineering. Das, B. M. Cengage Learning. ISBN: 978-0495668107
4. Geographic Information Systems and Science. Longley, P. A., Goodchild, M. F., Maguire, D. J., & Rhind, D. W. John Wiley & Sons. ISBN: 978-0470721446
5. Smart Cities: Big Data, Civic Hackers, and the Quest for a New Utopia. Townsend, A. M. W.W. Norton & Company. ISBN: 978-0393349786
6. Construction Project Management: A Practical Guide to Field Construction Management (. Sears, S. K., Sears, G. A., & Clough, R. H. John Wiley & Sons. ISBN: 978-1118745052

<table>
<tr><th>Course Title</th><th colspan="4">Advanced C Programming with Copilot Lab</th><th>Course Type</th><th>HC</th></tr>
<tr><td>Course Code</td><td>B25CI0202</td><td>Credits</td><td colspan="2">1</td><td>Class</td><td>II Semester</td></tr>
<tr><td rowspan="2"></td><td>LTP</td><td>Credits</td><td>Contact Hours</td><td>Work Load</td><td rowspan="2">Total Number of Classes Per Semester</td><td rowspan="2">Assessment in Weightage</td></tr>
<tr><td>Lecture</td><td>-</td><td>-</td><td>-</td></tr>
</table>

COURSE STRUCTURE	Tutorial	-	-	-	Theory	Practical	IA	SEE
	Practice	1	2	2				
	Total	1	2	2	-	28	**50%**	**50%**

Course Overview:

This course covers essential C programming concepts, including control structures, pointers, dynamic memory management, structures, and file handling. Students will gain practical experience by applying these skills to solve real-world problems. Additionally, the course introduces UNIX operating systems and shell scripting, focusing on basic commands and automation tasks. By the end, students will have a solid understanding of both C programming and UNIX scripting, preparing them for more advanced programming challenges.

COURSE OUTCOMES (COs)

On successful completion of this course; the student shall be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Develop modular and reusable code using functions (call by value and call by reference), structures, arrays and arrays of structures using illustrative programs and test-driven development.	1-3	2
CO2	Develop modular and reusable code using functions (call by value and call by reference), structures, arrays and arrays of structures using GitHub copilot using illustrative programs and test-driven development.	1-3	2
CO3	Develop modular and reusable code using pointers and dynamic memory allocation using illustrative programs and test-driven development.	1-3	2
CO4	Develop modular and reusable code using pointers and dynamic memory allocation using GitHub copilot using illustrative programs and test-driven development.	1-5	2
CO5	Develop a simple application with strings and file handling in C using illustrative programs and test-driven development	1-5	2,3
CO6	Develop a simple application with strings and file handling in C using GitHub copilot using illustrative programs and test-driven development	1, 5	2,3

BLOOM’S LEVELOF THE COURSE OUTCOMES

CO#	Bloom’s Level					
	Remember (L1)	**Understand (L2)**	**Apply (L3)**	**Analyze (L4)**	**Evaluate (L5)**	**Create (L6)**
CO1			√			
CO2			√			
CO3		√				
CO4			√			
CO5			√			
CO6				√		

COURSE ARTICULATION MATRIX:

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PS O1	PS O2	PS O3

CO1	3	3	3	3	3				3		3	3	2	3
CO2	2	2	2		3									
CO3	2	2	2											
CO4	2	2	3										2	
CO5	3	3	3		3				3	3	3	3	2	3
CO6	3	3	3	3	3				3		3	3	2	3

Note: 1-Low, 2-Medium, 3-High

Programs

Sl. No	Programs
1	Write programs using functions (call by value) to perform basic calculations using variables and expressions
2	Write programs using functions (call by reference) to perform basic calculations using variables and expressions
3	Write programs using functions (call by reference) to perform basic calculations using if-else, arrays and loops
4	Write programs using functions and structures to perform basic calculations using variables and expressions
5	Write programs using functions and structures to perform basic calculations using if-else and loops
6	Write programs using functions and array of structures to perform basic calculations using if-else and loops
7	Write programs using array of structures to perform CRUD operations like student records, employee records, library database, product details, examination results
8	Write programs using functions and pointers to perform basic pointer operations
9	Write programs using functions and pointers to perform dynamic memory allocation operations
10	Write programs using functions and structures to perform dynamic memory allocation operations
11	Write programs using functions to perform basic string operations
12	Write programs using functions and structures to perform basic file operations

TEXTBOOKS:

1. B.W. Kernighan & D.M. Ritchie, "C Programming Language", 2nd Edition, PRENTICE HALL SOFTWARE SERIES, 2005.
2. Herbert Schildt, “C: The Complete Reference”, 4th edition, TATA McGraw Hill, 2000.
3. B.S. Anami, S.A. Angadi and S. S. Manvi, “Computer Concepts and C Programming: A Holistic Approach”, second edition, PHI,2008.
4. Yashavant Kanetkar, “Let us C” , BPB , 19th edition, 2022

REFERENCE BOOKS:

1. Balaguruswamy,” Programming in ANSI C”, 4th edition, TATA MCGRAW Hill, 2008.
2. Donald Hearn, Pauline Baker,“ Computer Graphics C Version”, second edition, Pearson Education, 2004.
3. Jon Bentley, “Programming Pearls”, 2nd edition, PEARSON,2002.

JOURNALS/MAGAZINES:

1. https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=6294 (IEEE Journal/Magazine on IT Professional)
2. https://ieeexplore.ieee.org/document/1267572 (IEEE Computing in Science and Engineering)

SWAYAMNPTEL/MOOCs

1. https://online courses.nptel.ac.in/noc20_cs06/preview (Problem Solving through Programming in C)
2. https://www.edx.org/course/c-programming-getting-started (C Programming Getting started)
3. https://www.coursera.org/specializations/c-programming (Introduction to C programming)

<table>
<tr><td>Course Title</td><td colspan="4">Electronics and Digital Logic Lab</td><td colspan="2">Course Type</td><td colspan="2">HC</td></tr>
<tr><td>Course Code</td><td>B25EE0102</td><td>Credits</td><td colspan="2">1</td><td colspan="2">Class</td><td colspan="2">I semester</td></tr>
<tr><td rowspan="5">Course Structure</td><td>TLP</td><td>Credits</td><td>Contact Hours</td><td>Workload</td><td colspan="2" rowspan="2">13 weeks/ Semester</td><td colspan="2" rowspan="2">Assessment in Weightage</td></tr>
<tr><td>Theory</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>Tutorial</td><td>-</td><td>-</td><td>-</td><td rowspan="2">Theory</td><td rowspan="2">Practical</td><td rowspan="2">CIE</td><td rowspan="2">SEE</td></tr>
<tr><td>Practice</td><td>1</td><td>2</td><td>2</td></tr>
<tr><td>Total</td><td>1</td><td>2</td><td>2</td><td>-</td><td>28</td><td>50%</td><td>50%</td></tr>
</table>

COURSE OVERVIEW

This course covers basic concepts of Electrical Engineering. The course introduces the working of analog components and helps in understanding basics in digital electronics by applying the knowledge of logic gates and learning the applications of diodes. The course provides foundation on designing and implementation of analog and logic circuits. This course develops students' ability to understand and design the basic building blocks of modern digital and analog systems and provides them with a fundamental knowledge for complicated digital and analog hardware design

COURSE OBJECTIVE (S):

The objectives of this course are to:

1. Discuss the applications of diode in rectifiers circuits.
2. Discuss the applications of diode in wave shaping circuits.
3. Illustrate Boolean laws and systematic techniques for minimization of expressions.
4. Demonstrate the methods for simplifying Boolean expressions.
5. Familiarize the commonly used terms like min-term, max-term, canonical expression, SOP, POS etc.
6. Introduce the Basic concepts of combinational and sequential logic.
7. Present real-world examples for making the learners attuned to Logic concepts.
8. Highlight the formal procedures for the analysis and design of combinational circuits and sequential circuits.
9. Introduce the concept of memories, programmable logic devices and digital ICs

COURSE OUTCOMES (COs)

After the completion of the course, the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Analyze the use of diodes working principle of halfwave rectifier and fullwave rectifier.	1,2,3,4	1,2,3
CO2	Analyze the use of diodes in clipping ,clamping circuits and working principle.	1,2,3,,5	1,2,3

CO3	Implement Demorgan's theorem using logic gates and verify their truth table.	1,2,3,4,5	1,2,3
CO4	Implement Half adder, Full adder, Half subtractor, and Full subtractor circuits using logic gates and verify their truth table.	1,2,3,4	1,2,3
CO5	Implement Multiplexer and Demultiplexer using logic gates and verify its functionality.	1,2,3,4	1,2,3
CO6	Construct and test JK master-slave, T, and D flip-flops, and verify their truth tables.	1,4,5,	1,2,3

BLOOM'S LEVEL OF THE COURSE OUTCOMES

CO#	Bloom's Level					
	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1				√		
CO2			√			
CO3			√			
CO4				√		
CO5				√		
CO6			√			

COURSE ARTICULATION MATRIX

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	1	1	2								3	1	3
CO2	3	2	3		2							3	1	3
CO3	3	1	2	1	2							3	1	3
CO4	3	1	1	2								3	1	3
CO5	3	1	1	2								3	2	3
CO6	3			2	2							3	2	3

Note: 1-Low, 2-Medium, 3-High

No	Title of the Experiment	Tools and Techniques	Expected Skill /Ability
1	Verify and realization Halfwave-rectifier	Diode, power supply, connector board, wires and CRO	Design and circuit debugging
2	Verify and realization Fullwave rectifier.	Diode, power supply, connector board, wires and CRO	Design and circuit debugging
3	Realization of working principal of positive & Negative clipper Circuits.	JSCircuit Software	Design and circuit debugging
4.	Realization of working principal of positive and negative clamper circuits.	JSCircuit Software	Design and circuit debugging
5.	To verify truth table of AND, OR, NOT gate and universal gates	ICs, Trainer kit and patch cords	Design and circuit debugging
6.	To Verify Demorgan's Theorem for 2 variables expressions	ICs, Trainer kit and patch cords	Design and circuit debugging
7.	Realization of Half Adder and Full Adder using logic gates	ICs, Trainer kit and patch cords	Design and circuit debugging
8.	Realization of Half Substractor and Full Substractor using logic gates	ICs, Trainer kit and patch cords	Design and circuit debugging
9.	Design and implementation of Multiplexer and Demultiplexer	ICs, Trainer kit and patch cords	Design and circuit debugging
10.	Test and verify the truth-table of the following flip-flops using IC7476: a. JK flip-flop b. D flip-flop c. T flip-flop	ICs, Trainer kit and patch cords	Design and circuit debugging

TEXT BOOKS:

1. R.P. Jain, Modern Digital Electronics, McGraw Hill
2. Thomas L. Floyd, Electronic Devices and Circuit Theory, Pearson
3. Ramakant A. Gayakwad, Op-Amps and Linear Integrated Circuits, PHI Learning

REFERENCE BOOKS:

1. Millman & Halkias, Electronic Devices and Circuits, McGraw Hill
2. Tocci & Widmer, Digital Systems: Principles and Applications, Pearson
3. Sedra & Smith, Microelectronic Circuits, Oxford University Press

JOURNALS / MAGAZINES:

1. IEEE Transactions on Circuits and Systems
2. International Journal of Electronics and Communication Engineering
3. Electronics For You (EFY) magazine
4. IETE Journal of Research

SWAYAM / NPTEL / MOOCs:

1. NPTEL – Basic Electronics by Prof. T.K. Basu, IIT Kharagpur
2. NPTEL – Digital Electronics by Prof. S. Srinivasan, IIT Madras
3. SWAYAM – Digital Logic Design by Prof. D. Roy Choudhury, JNU Delhi
4. Coursera – Introduction to Electronics (Georgia Tech)
5. edX – Circuits and Electronics (MITx)

Course Title	**Python for Data Science Lab**				**Course Type**			
Course Code	**B25CS0102**	**Credits**	**1**		**Class**		**II Semester**	
Course Structure	TLP	Credits	Contact Hours	Work Load	Total Number of Classes Per Semester		Assessment in Weightage	
	Theory	-	-	-				
	Practice	1	2	2	Theory	Practical	CIE	SEE
	-	-	-	-				
	Total	**1**	**2**	**2**	**-**	**26**	**25**	**25**

COURSE OVERVIEW:

The *Python for Data Science Lab* offers hands-on practice in programming, data manipulation, visualization, and exploratory analysis using Python libraries like NumPy, Pandas, Matplotlib, Seaborn, and scikit-learn. Culminating in a mini-project, it equips students with practical skills to handle end-to-end data workflows and solve real-world problems.

COURSE OBJECTIVE (S):

The objectives of this course are to:

1. Provide hands-on experience in writing and executing Python programs with fundamental constructs.
2. Familiarize students with core data structures and modular programming techniques in Python.
3. Enable the handling of structured datasets through file I/O, NumPy arrays, and Pandas data frames.
4. Develop proficiency in cleaning, manipulating, and preparing datasets for analysis.
5. Train students in creating effective data visualizations using Matplotlib and Seaborn.
6. Introduce the process of exploratory data analysis (EDA) and predictive modeling using scikit-learn.
7. Apply Python programming to solve real-world data science problems through a structured mini-project.

COURSE OUTCOMES (COs)

After the completion of the course, the student will be able to:

CO #	**Course Outcomes**	**POs Addressed**	**PSOs Addressed**

CO1	Write and execute basic Python programs demonstrating variables, arithmetic operations, input/output, and control structures.	PO1, PO2, PO3, PO10	PSO1
CO2	Develop modular code using functions and effectively manipulate built-in data structures (lists, tuples, sets, dictionaries).	PO1, PO2, PO3, PO10	PSO1
CO3	Perform file handling and structured data processing from text and CSV files.	PO1, PO2, PO3, PO4, PO10	PSO1, PSO2, PSO3
CO4	Apply NumPy and Pandas to clean, transform, and manipulate datasets efficiently.	PO1, PO2, PO3, PO4, PO10	PSO1, PSO2, PSO3
CO5	Create and customize data visualizations using Matplotlib and Seaborn.	PO1, PO2, PO3, PO4, PO5, PO10	PSO1, PSO2, PSO3
CO6	Conduct exploratory data analysis (EDA) on given datasets.	PO1, PO2, PO3, PO4, PO5, PO9, PO10	PSO1, PSO2, PSO3
CO7	Implement simple predictive models using scikit-learn.	PO1, PO2, PO3, PO4, PO5, PO6, PO9, PO10, PO11	PSO1, PSO2, PSO3
CO8	Design and execute a mini-project that applies Python to an end-to-	PO1, PO2, PO3,	PSO1, PSO2,

CO #	Course Outcomes	POs Addressed	PSOs Addressed
	end data science workflow.	PO4, PO5, PO6, PO7, PO9, PO10, PO11	PSO3

COURSE ARTICULATION MATRIX

Course Outcomes (COs)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	-	-	-	-	1	-	3	-	-
CO2	3	3	2	-	-	-	-	-	-	1	-	3	-	-
CO3	2	3	2	1	-	-	-	-	-	1	-	3	2	2
CO4	2	3	2	2	-	-	-	-	-	1	-	3	3	2
CO5	2	2	2	2	1	-	-	-	-	2	-	3	3	2
CO6	2	3	3	3	2	-	-	-	1	2	-	3	3	2
CO7	2	3	3	3	2	2	-	-	1	2	-	3	3	2
CO8	2	3	3	3	3	2	1	-	2	3	1	3	3	2

Note:1-Low,2-Medium,3-High

PRACTICE:

Sl. No	Title of the Experiment	Tools and Techniques	Expected Skills / Ability
1	Writing basic Python programs (Hello World, arithmetic operations)	Jupyter Notebook / Google Colab, Python 3.x	Understand basic syntax, variables, operators, and execution of simple programs.
2	Implementing control structures and functions in Python	Jupyter Notebook / Google Colab, Python IDE (VS Code)	Apply decision-making (if/else), looping constructs, and modular programming using functions.
3	Manipulating lists, tuples, sets, and dictionaries	Python built-in data structures, IDE/Colab	Gain proficiency in using and modifying Python's core data structures for problem-solving.
4	Reading and writing data from text and CSV files	Python I/O functions, Pandas	Handle file operations, perform basic data input/output processing with structured files.
5	Performing array operations using NumPy	NumPy library in Jupyter / Colab	Develop ability to create, reshape, and compute with numerical arrays.
6	Data manipulation and cleaning using Pandas	Pandas library (Series & DataFrames)	Perform dataset cleaning, filtering, aggregation, and basic preprocessing

Sl. No	Title of the Experiment	Tools and Techniques	Expected Skills / Ability
			tasks.
7	Creating various plots with Matplotlib and Seaborn	Matplotlib, Seaborn libraries	Create line, bar, scatter, pie, box plots with customization for visualization.
8	Exploratory Data Analysis (EDA) on sample datasets	Pandas, Seaborn, Matplotlib	Perform descriptive statistical analysis, visualize patterns, detect outliers.
9	Building a simple predictive model with scikit-learn	scikit-learn (Linear Regression, Classification)	Develop skills to train, test and evaluate simple ML models using Python.
10	Mini project: Analyzing a real-world dataset end-to-end	Jupyter Notebook, Pandas, NumPy, Seaborn, scikit-learn	Integrate skills in data collection, cleaning, analysis, visualization, and model building.

ourse title	**Finance and Management**				**Course type**		**FC**	
Course code	B24EN0102	Credits	1		**Class**		**I Semester**	
COURSE STRUCTURE	LTP	Credits	Contact hours	Work load	Total Number of classes per semester		Assessment in weightage	
	Lecture	1	1	1				
	Tutorial	-	-	-	Theory	Practical	CIE	SEE
	Practice	-	-	-				
	Total	**1**	**1**	**1**	**14**	**-**	**50%**	**50%**

Course overview

This course introduces the fundamentals of finance and management principles essential for understanding business operations and decision-making processes. It covers key concepts in finance such as the time value of money, financial markets, and institutions, along with principles of management including planning, organizing, leading, and controlling.

Course objectives

1. Understand the concept of finance and its various applications in both personal and business contexts.
2. Evaluate various financial instruments, markets, and investment opportunities to make informed investment decisions.
3. Develop the ability to analyze financial situations using time value of money principles.
4. Apply management theories and techniques to effectively lead teams, manage conflicts, and enhance organizational performance.
5. Examine Henri Fayol's fourteen principles of management and their relevance to contemporary organizational practices.

COURSE OUTCOMES (COs):

On successful completion of this course, the student shall be able to:

CO#	Course outcomes	POs	PSo
C01	Apply foundational financial principles to everyday decisions—including budgeting, saving, investing, and spending—to enhance personal financial well-being and long-term stability.	6,7,8,11	1
C02	Analyze how various financial intermediaries—such as banks, investment banks, and exchanges—facilitate fund allocation, underpin market operations, and drive economic growth.	6,7,8,11	1
CO3	Apply present and future value calculations to inform business and personal finance decisions such as saving, borrowing, investing, and planning.	6,7,8,11	1
CO4	Identify the functions of management: planning, organizing, leading, and controlling.	7,8,10,11	1
C05	Analyze key motivation theories and apply them to understand and influence employee behavior in organizational contexts.	7,8,10,11	1
CO6	Develop effective communication skills and decision-making abilities essential for managerial roles, including interpersonal communication, conflict resolution, and problem-solving.	7,8,10,11	1

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

	Remember (L1)	Understand (L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1	✓	✓	✓			
CO2	✓	✓	✓			
CO3	✓	✓	✓			
CO4	✓	✓	✓			
CO5	✓	✓	✓			
CO6	✓	✓	✓			

COURSE ARTICULATION MATRIX:

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1						1	1	1			1	1		
CO2						1	1	1			1	1		
CO3						1	1	1			1	1		
CO4							1	1		1	1	1		
CO5							1	1		1	1	1		
CO6							1	1		1	1	1		

Note: 1-Low, 2-Medium, 3-High

COURSE CONTENTS:

THEORY:

CONTENTS
Introduction to Finance Overview of Finance Definition and Scope of Finance Importance of Finance in Business and Personal Decision Making Goals of Financial Management Financial Markets and Institutions: Personal Financial Planning. Reading Understanding financial statements. Time Value of Money Concepts: Future Value, Present Value, Discounting Calculating Future Value and Present Value of Cash Flows Applications of Time Value of Money in Financial Decision Making Financial Markets and Institutions Types of Financial Markets: Money Market, Capital Market, Primary Market, Secondary Market Financial Instruments: Stocks, Bonds, Derivatives Role of Central Banks and Regulatory Bodies in Financial markets. **Principles of Management** Management: Introduction- Meaning, nature and characteristics of management Definition of management Evolution of management thought Scope and Functions of management (Planning, Organizing, Leading, and Controlling). Management as a science art or profession. Management and administration. Levels of management (Top, Middle, Front-line). Principles of management- Fayols principles of management. Taylor's Scientific Management Theory- Taylor's Principles of Management, Maslow's theory of Hierarchy of Human Needs, Douglas McGregor's Theory X and Theory Y- Hertzberg Two Factor Theory of Motivation - Leadership Styles, Social responsibilities of Management. Porter's Five Force Model, BCG Matrix and PESTLE analysis.

Self-Learning Component:

1. Study Future Value, Present Value, and Discounting concepts through online tutorials or videos.
2. Investigate examples of companies demonstrating social responsibility and their impact on stakeholders.
3. Explore how each financial instrument is traded and valued in the market.
4. Reflect on personal leadership qualities and areas for development.

TEXT BOOKS:

1. "TAXMANN's Financial Management" by Ravi M. Kishore(2020) - Publisher: Taxmann Publications Pvt. Ltd. ISBN: 978-9389702385
2. Management Appannaiah, H. R. & Reddy, P. Second Revised Edition Himalaya Publishing House.

REFERENCE BOOKS

1. "Financial Management" Prasanna Chandra –Publisher: McGraw-Hill Education.(2020)ISBN-13: 978-9389143895
2. Management Richard L. Daft Publisher: Cengage Learning.

<table>
<tr><td>Course Title</td><td colspan="4">Engineering Exploration</td><td colspan="2">Course Type</td><td colspan="2">ETC</td></tr>
<tr><td>Course Code</td><td>B24CIET01</td><td>Credits</td><td colspan="2">1</td><td colspan="2">Class</td><td colspan="2">II semester</td></tr>
<tr><td rowspan="5">COURSE STRUCTURE</td><td>LTP</td><td>Credits</td><td>Contact Hours</td><td>Work Load</td><td colspan="2" rowspan="2">Total Number of Classes Per Semester</td><td colspan="2" rowspan="2">Assessment Weightage</td></tr>
<tr><td>Lecture</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>Tutorial</td><td>-</td><td>-</td><td>-</td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><td>Practice</td><td>-</td><td>-</td><td>-</td><td>Theory</td><td>Practical</td><td>CIE</td><td>SEE</td></tr>
<tr><td>Total</td><td>1</td><td>1</td><td>1</td><td>14</td><td>-</td><td>50%</td><td>50%</td></tr>
</table>

COURSE OVERVIEW:

This course provides a comprehensive introduction to the field of engineering, aimed at equipping students with the foundational knowledge and skills necessary to pursue a career in this dynamic and impactful discipline. The curriculum is designed to cover a broad spectrum of topics, from the historical and contemporary contributions of engineering to the principles and practices that define the profession today, with a particular emphasis on the Indian context.

COURSE OBJECTIVE (S):

The objectives of this course are to:

1. Introduce in the field of engineering, emphasizing the role of engineers as problem solvers and innovators.
2. Familiarize with the global and Indian engineering industries, highlighting various career opportunities.
3. Develop an engineering mindset by analyzing the design and functioning of common engineering artifacts.
4. Enhance ability to work effectively in multidisciplinary teams, utilizing problem-solving skills, project management concepts, and effective communication of engineering ideas through written reports and presentations.

COURSE OUTCOMES (COs)

After the completion of the course, the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO1	Explain what engineering is, the role of an engineer as a problem solver and appreciate the history and impact of significant engineering achievements especially in the Indian context.	1,2	1
CO2	Be familiar with the global and Indian engineering industry and the various career opportunities it offers.	1,2	1
CO3	Demonstrate Engineering Mindset by analyzing design and working of common engineering artefacts and Gain practical insights through case studies, product dissection, and industry interactions.	1,4	2
CO4	Apply the engineering design process and principles to identify needs, define problems, and develop solutions	2,3	1
CO5	Demonstrate the ability to work effectively in multidisciplinary teams, Use problem Solving Skills, apply project management concepts, and communicate engineering ideas through written reports and presentations.	5,9,10	3
CO6	Recognize the importance of professional ethics, sustainability, and the societal impact of engineering solutions, and incorporate these considerations into the design process.	6,7,8	3

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

CO#	Bloom's Level					
	Remember(L1)	Understand(L2)	Apply (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO1						
CO2		√				
CO3		√				
CO4			√			
CO5						
CO6				√		

COURSE ARTICULATION MATRIX:

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3										3		
CO2	3	3										3		
CO3	3		3										3	
CO4		3	3									3		
CO5					3				3	3				3
CO6						2	3	3						3

Note: 1-Low, 2-Medium, 3-High

COURSE CONTENT

COURSE CONTENTS
Definition of engineering, its scope, and various disciplines, Understanding the engineer's role in identifying, defining, and solving problems, Examples of engineering problem-solving in everyday life and industry. Overview of engineering through different historical periods, Key milestones and developments in engineering practices, Case studies on significant engineering achievements and their societal impacts, Overview of historical engineering achievements in India (e.g., ancient water management systems, Indus Valley civilization), Modern engineering marvels in India (e.g., ISRO's achievements, infrastructure projects). Present-day engineering issues (e.g., sustainability, climate change, urbanization), Role of engineers in addressing these challenges, Exploration of emerging fields (e.g., artificial intelligence, biotechnology, renewable energy), Career opportunities global and Indian engineering industry.

Definition and characteristics of an engineering mindset, Importance of critical thinking, problem-solving, and creativity in engineering, Design thinking process, Application of design principles in engineering artefacts, Common engineering product for its design principles and functionality.

Case Study Analysis: Successful Engineering Designs, Examination of case studies (e.g., bridges, consumer electronics, medical devices), Factors contributing to the success of these designs, Product Dissection, Techniques and importance of product dissection in understanding engineering principles, Hands-on dissection of a chosen engineering product (Mobile or Two Wheller or Laptop).

Importance of Multidisciplinary Teams in Engineering Projects, Team Dynamics and Collaboration, Engineering Problem-Solving Process, Application of problem-solving frameworks (e.g., TRIZ, Six Sigma), Project Management Concepts in Engineering, Visual Communication in Engineering, diagrams, graphs, and visual aids in technical reports and presentations, Tools for creating effective visual representations of engineering concepts.

Improvements for an existing engineering artefact, considering ethical and sustainability aspects.

TEXTBOOKS:

1. "Engineering Fundamentals: An Introduction to Engineering" by Saeed Moaveni
2. "Introduction to Engineering" by Paul H. Wright
3. "The Engineering Design Process" by Atila Ertas and Jesse C. Jones
4. "Ethics in Engineering" by Mike W. Martin and Roland Schinzinger
5. Web Based Resources and E-books:
6. "Engineering Your Future: A Brief Introduction to Engineering" by William Oakes, Les Leone, and Craig Gunn

<table>
<tr><td>Course Title</td><td colspan="4">AI Foundations for Engineers</td><td>Course Type</td><td colspan="2">ETC</td></tr>
<tr><td>Course Code</td><td>B25CSET01</td><td>Credits</td><td colspan="2">1</td><td>Class</td><td colspan="2">II Semester</td></tr>
<tr><td rowspan="5">Course Structure</td><td>LTP</td><td>Credits</td><td>Contact Hours</td><td>Work Load</td><td rowspan="2">Total Number of Classes Per Semester</td><td colspan="2" rowspan="2">Assessment in Weightage</td></tr>
<tr><td>Lecture</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>Tutorial</td><td>-</td><td>-</td><td>-</td><td rowspan="2">Theory</td><td rowspan="2">IA</td><td rowspan="2">SEE</td></tr>
<tr><td>Practical</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>Total</td><td>1</td><td>1</td><td>1</td><td>14</td><td>50%</td><td>50%</td></tr>
</table>

Pre-requisite

NA

Overview of Course

This course introduces engineering students to fundamental concepts of Artificial Intelligence with emphasis on experimental learning and project-based skill development. Students will explore AI applications across engineering domains, implement basic AI algorithms, and develop practical solutions through hands-on projects. The course emphasizes critical thinking, problem-solving, and ethical considerations in AI deployment, preparing students for advanced AI courses and real-world applications in their respective engineering disciplines.

Course Objectives

1. To enable students to demonstrate proficiency in Python programming fundamentals, including syntax, variables, control structures, and core data structures (lists, tuples, sets, dictionaries) for data science applications.
2. To develop students' ability to perform data wrangling, cleaning, and analysis using industry-standard Python libraries such as Pandas and NumPy for handling structured datasets effectively.
3. To equip students with skills to create meaningful data visualizations using Matplotlib and Seaborn libraries, enabling them to customize, evaluate, and present data insights in a clear and professional manner.
4. To foster practical problem-solving abilities through hands-on mini-projects that address real-world data science challenges, preparing students for advanced courses and interdisciplinary projects in the field of Data Science.

Course Outcomes

Upon completion, students will be able to:

CO Code	Course Outcome Statement	PO's	PSO's
CO1	Explain fundamental concepts of AI, its evolution, and applications in various engineering domains with ethical considerations.	1,2,5,6,8,10	1,2
CO2	Implement basic search algorithms and optimization techniques for solving engineering problems using modern programming tools.	1,2,3,4,5,9,10,11	1,2,3
CO3	Analyze and apply knowledge representation techniques and reasoning methods for engineering applications.	1,2,3,4,5,8,9,10,11	1,2,3
CO4	Design simple machine learning models and evaluate their performance for engineering data analysis and prediction tasks.	1,2,3,4,5,7,8,9,10,11	1,2,3
CO5	Develop project-based solutions integrating AI techniques to address real-world engineering challenges with sustainability focus.	1,2,3,4,5,6,7,8,9,10,11	1,2,3
CO6	Demonstrate effective communication and teamwork skills in presenting AI-based engineering solutions and their societal impact.	1,2,3,4,5,6,7,8,9,10,11	1,2,3

BLOOM'S LEVEL OF THE COURSE OUTCOMES:

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2			1	1		3		1		3	2	
CO2	3	3	2	2	3				1	1	1	3	3	2
CO3	3	3	3	2	3			1	1	1	1	3	3	2
CO4	2	3	3	3	3		1	1	2	2	2	3	3	3
CO5	2	3	3	3	3	3	3	3	3	3	3	3	3	3
CO6	1	2	2	2	2	1	1	3	3	3	3	2	2	2

COURSE CONTENT:

Introduction to Artificial Intelligence

Definition, scope, and history of AI, AI applications in Civil, Mechanical, Electrical, Electronics, and Computer Engineering, Types of AI: Narrow AI, General AI, and Superintelligence, AI vs Traditional Programming approaches, Ethical implications and societal impact of AI in engineering, Current trends: Industry 4.0, Smart Cities, IoT integration, Case study analysis on AI applications in different engineering domains, Interactive exploration of AI tools and platforms (ChatGPT, Google AI, etc.), Group discussions on ethical dilemmas in AI implementation

Problem Solving and Search Techniques

Problem formulation and state space representation, Uninformed search: Breadth-First Search (BFS), Depth-First Search (DFS), Informed search: Greedy Best-First Search, A* algorithm, Optimization techniques: Hill Climbing, Genetic Algorithms, Constraint Satisfaction Problems in engineering contexts

Knowledge Representation and Reasoning

Knowledge representation schemes in engineering systems, Rule-based systems and expert systems, Propositional and predicate logic for engineering applications, Semantic networks and ontologies, Uncertainty handling and probabilistic reasoning, Bayesian networks for engineering diagnosis.

Machine Learning Foundations

Introduction to Machine Learning paradigms**,** Supervised learning: Classification and Regression for engineering data**,** Unsupervised learning: Clustering for pattern recognition**,** Key algorithms: Linear Regression, Decision Trees, k-Means clustering**,** Model evaluation techniques and cross-validation**,** Feature selection and engineering data preprocessing

AI Project Development and Applications

Project lifecycle for AI-based engineering solutions, Integration of AI with IoT and embedded systems, Sustainable AI practices in engineering applications, AI safety and reliability in critical engineering systems, Future trends: Edge AI, Quantum Computing, Green AI.

Textbooks:

[T1] Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Pearson, 2020

[T2] Tom Mitchell, Machine Learning, McGraw Hill, 2017

References:

[R1] Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition, O'Reilly, 2019

[R2] Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016

[R3] Coursera: "AI for Everyone" by Andrew Ng

[R4] edX: "Introduction to Artificial Intelligence" by Microsoft

[R5] MIT OpenCourseWare: "Introduction to Machine Learning"

REVA UNIVERSITY
Bengaluru, India

Rukmini Knowledge Park, Kattigenahalli
Yelahanka, Bengaluru - 560 064
Karnataka, India.

Ph: +91- 90211 90211, +91 80 4696 6966
E-mail: admissions@reva.edu.in

www.reva.edu.in

Follow us on

/revauniversity_official

/REVAUniversity

/revauniversity_official

/@revauniversity_official

/REVA University

reva.edu.in

+91 90211 90211